

利用 Lean/Mathlib 审计数学论文中低声望高风险技术步骤的候选方向调研

Formal audit of low-glory high-risk mathematical infrastructure in research papers

数学文献调研与 Lean/形式化数学顾问报告

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摘要

本报告调研一种非传统形式化数学工作流：不以完整形式化整篇论文为目标，而是把真实研究论文中被写成 *standard, routine, by base change, by descent, by functoriality, the diagram commutes* 的技术步骤抽成可审计、可复用、可局部形式化的任务。调研重点围绕 Lean 4、Mathlib、代数几何与 Jacobian formalization 基础设施，特别关注 Picard/Albanese/Jacobian、Hilbert/Quot/flattening、cohomology and base change、descent/base change、abelian variety rigidity、heavy computation 以及真实数学 benchmark。

目录

1	Executive Summary / 执行摘要	7
1.1	核心判断	7
1.2	Lean/Mathlib 贴合度的现实边界	8
1.3	最推荐的 5 个 pilot 方向	9
1.4	本报告的结论一句话	9

2 核心价值定位	9
2.1 这不是完整形式化论文	9
2.2 六类价值及优先级	10
2.2.1 重计算 verify outsourcing	10
2.2.2 Hidden hypothesis audit	10
2.2.3 Black-box theorem interface specification	11
2.2.4 Reusable formal infrastructure extraction	11
2.2.5 Real-math benchmark for AI theorem proving	12
2.2.6 Proof audit / technical due diligence for mathematical papers . .	12
2.3 本方向的输出形态	12
3 论文/主题候选总览	13
3.1 筛选原则	13
3.2 优先级粗分	20
4 分方向详细分析	20
4.1 A. Picard / Albanese / Jacobian / line bundles	20
4.1.1 数学背景	20
4.1.2 常见 proof smell	20
4.1.3 推荐候选论文	21
4.1.4 Lean/Mathlib 贴合度	21
4.1.5 依赖爆炸风险	22
4.1.6 最适合抽取的 Lean-style task	22
4.2 B. Hilbert / Quot / flattening / representability	22
4.2.1 数学背景	22
4.2.2 常见 proof smell	23
4.2.3 推荐候选论文	23

4.2.4	Lean/Mathlib 贴合度	23
4.2.5	依赖爆炸风险	24
4.2.6	最适合抽取的 Lean-style task	24
4.3	C. Cohomology and base change / higher direct image	24
4.3.1	数学背景	24
4.3.2	常见 proof smell	24
4.3.3	推荐候选论文/资料	25
4.3.4	Lean/Mathlib 贴合度	25
4.3.5	依赖爆炸风险	25
4.3.6	最适合抽取的 Lean-style task	25
4.4	D. Descent / base change / algebraic closure arguments	26
4.4.1	数学背景	26
4.4.2	常见 proof smell	26
4.4.3	推荐候选论文	26
4.4.4	Lean/Mathlib 贴合度	27
4.4.5	依赖爆炸风险	27
4.4.6	最适合抽取的 Lean-style task	27
4.5	E. Abelian variety rigidity / rational maps / theorem of the cube	28
4.5.1	数学背景	28
4.5.2	常见 proof smell	28
4.5.3	推荐资料/论文类型	28
4.5.4	Lean/Mathlib 贴合度	29
4.5.5	依赖爆炸风险	29
4.5.6	最适合抽取的 Lean-style task	29
4.6	F. Heavy computation / intersection theory / spectral sequence book-keeping	29

4.6.1	数学背景	29
4.6.2	推荐候选论文	30
4.6.3	常见 proof smell	30
4.6.4	Lean/Mathlib 贴合度	30
4.6.5	依赖爆炸风险	30
4.6.6	最适合抽取的 Lean-style task	31
4.7	G. Real-math Lean benchmark extraction	31
4.7.1	数学背景	31
4.7.2	常见 benchmark 缺口	31
4.7.3	推荐候选资料	31
4.7.4	Lean/Mathlib 贴合度	32
4.7.5	最适合抽取的 Lean-style task	32
5	候选论文卡片	32
5.1	P01. <i>The Picard scheme</i>	32
5.2	P02. <i>Compactifying the Picard scheme</i>	33
5.3	P03. <i>Compactifying the relative Jacobian over families of reduced curves</i> .	34
5.4	P04. <i>A compactification of the universal Picard variety over the moduli space of stable curves</i>	35
5.5	P05. <i>Néron models and compactified Picard schemes over the moduli stack of stable curves</i>	36
5.6	P06. <i>Compactified Picard stacks over the moduli stack of stable curves with marked points</i>	37
5.7	P07. <i>Autoduality of compactified Jacobians for curves with plane singularities</i>	38
5.8	P08. <i>Fourier-Mukai and autoduality for compactified Jacobians I/II</i>	39
5.9	P09. <i>Construction of Hilbert and Quot Schemes</i>	40
5.10	P10. <i>Quot Functors for Deligne-Mumford Stacks</i>	41

5.11	P11. <i>Cohomology and base change for algebraic stacks</i>	42
5.12	P12. <i>Moduli of complexes on a proper morphism</i>	43
5.13	P13. <i>Versal deformations and algebraic stacks</i>	44
5.14	P16. <i>On the moduli of vector bundles on an algebraic surface</i>	45
5.15	P18. <i>A complete answer to Albanese base change for incomplete varieties</i>	46
5.16	P19. <i>Para-abelian varieties and Albanese maps</i>	47
5.17	P20. <i>Albanese maps for open algebraic spaces</i>	48
5.18	P22. <i>Chern classes of tautological sheaves on Hilbert schemes of points on surfaces</i>	49
6	候选 formal audit tasks 清单	50
6.1	任务粒度原则	50
6.2	首批最推荐的 10 个 tasks	53
6.3	示例 Lean-style statement 草案	53
6.3.1	Picard quotient functoriality	54
6.3.2	Descent of equality	54
6.3.3	Cohomology/base change theorem registry item	54
6.3.4	Heavy computation benchmark	55
7	优先级路线图	55
7.1	阶段一：快速调研和筛选	55
7.2	阶段二：深度阅读和局部 statement 提取	56
7.3	阶段三：形式化试点	56
7.4	建议的 8 周执行节奏	57
7.5	阶段性产物的文件组织建议	58
8	风险与反例	58
8.1	需要完整重建 EGA/SGA 的论文	58

- 8.2 Conceptual insight 大于 technical glue 的论文 59
- 8.3 Analytic / transcendental methods 太多 59
- 8.4 结论重要但局部不可切分 59
- 8.5 依赖大量尚不存在基础设施且复用价值不明确 60
- 8.6 隐藏假设审计的假阳性风险 60
- 8.7 Lean 当前能力被高估的风险 60
- 9 最终推荐 61
 - 9.1 最推荐优先探索的 5 个主题方向 61
 - 9.1.1 1. Relative Picard functor and line-bundle coherence 61
 - 9.1.2 2. Descent of equality and algebraic-closure arguments 61
 - 9.1.3 3. Cohomology and base change theorem registry 62
 - 9.1.4 4. Hilbert/Quot/flattening as black-box interfaces 62
 - 9.1.5 5. Heavy-computation microbenchmarks 63
 - 9.2 Recommended next-step plan 63
 - 9.3 最终一句推荐 64
- 10 附录：三轮调研与文稿修订日志 64
- 11 附录：主要资料与链接 66

1 Executive Summary / 执行摘要

1.1 核心判断

这个方向的核心价值不在于“证明某篇论文错了”，也不在于把一篇现代代数几何论文端到端形式化。更可行、也更有研究价值的目标是：把论文中作者和读者都不愿逐项展开的 *low-glory high-risk infrastructure* 转化为一组可审计的 *formal audit tasks*。这些 tasks 的粒度应小到能够成为 Lean statement、局部 proof sketch、blueprint 或 CI 检查项；同时又必须来自真实研究论文，而不是教科书练习。

本次调研后的优先级判断如下。

1. **Picard / line bundle functoriality / relative Picard interface** 是第一优先级。它与 Jacobian formalization 的现有路线最贴合，且能把 *Picard functor*, *Poincare bundle*, *Abel-Jacobi map*, Pic^0 , *base change* 等高频黑箱拆成可复用接口。Kleiman 的 *The Picard scheme*、Stacks Project 的 *Picard functor* 章节、Altman–Kleiman、Esteves、Arinkin、Caporaso/Melo 相关文献都适合做候选池。
2. **Descent / base change / algebraic closure arguments** 是第二优先级。这是最典型的 hidden hypothesis source：论文中“we may assume the base field is algebraically closed”之后，是否真的能 descend objects、morphisms、equalities、universal properties、connected components 和 numerical conditions，往往没有逐项写出。Achter–Casalaina–Martin–Vial 的 Albanese base change 论文和 Laurent–Schröer、Schröer 的 para-abelian/Albanese 文献是非常好的现代样本。
3. **Cohomology and base change** 应作为黑箱 **theorem registry** 与 **naturality benchmark**。完整形式化高阶 direct image 与 cohomology/base change 目前成本很高，但审计报告可以先把每次引用所需的 hypotheses 记录为 Lean-style theorem signatures。Conrad notes、Stacks Project、Hall 的 algebraic stacks 版本可作为来源。
4. **Hilbert / Quot / flattening stratification** 应先做 **interface specification**，不宜直接冲 **full proof**。Nitsure 的讲义是极好的“黑箱接口说明书”；Olsson–Starr 和 Lieblich 则展示了 stack/derived 版本会迅速 dependency explosion。早期 pilot 应只抽取 Quot functor base-change、flatness locus、Hilbert polynomial constancy 等局部 tasks。
5. **Heavy computation** 方向适合作为 **AI theorem proving benchmark**。

Ellingsrud–Strømme、Lehn、Ran 等文献中的 intersection number、Chern class、generating function、spectral sequence bookkeeping 可以从几何语境中剥离出多项式/组合/graded algebra 计算，成为 Lean agents 的非玩具 benchmark。

1.2 Lean/Mathlib 贴合度的现实边界

截至 2026-06-16 的公开文档显示, Mathlib 已有 Scheme、大量 morphism properties、AG sites、modules over schemes、Kähler differentials、ring-level Picard group、Proj construction 与 derived categories 相关基础; 这些足以支撑若干 *interface, naturality, descent of equality, algebraic calculation* 任务的局部形式化。但完整的 Picard scheme、relative Picard functor representability、Hilbert/Quot scheme、sheaf cohomology/higher direct image、intersection theory/Chow rings、abelian varieties/Jacobians 的主体理论仍需要项目级建设或大量新 Mathlib 基础设施。因此, 本报告使用四级 readiness 标记:

1. Mathlib-supported / 可立即尝试

抽象代数、category theory、basic scheme/morphism property、fpqc-local property、polynomial/formal power series 计算、某些 morphism equality descent。

2. Project scaffold / 项目已有雏形或路线图

line bundles as sheaves、relative Picard functor、Abel-Jacobi/Jacobian blueprint、pullback-tensor coherence、Poincare bundle normalization。

3. Major infrastructure / 需要大量新建

sheaf cohomology/higher direct image、Hilbert/Quot scheme、flattening stratification、Picard scheme representability、proper flat families of curves 的完整理论。

4. Not yet suitable / 暂不适合

Artin representability full proof、derived algebraic stacks、Fourier-Mukai equivalence on compactified Jacobians、full GIT construction、full Chow/intersection theory。

1.3 最推荐的 5 个 pilot 方向

1. **Picard quotient functoriality + pullback-tensor compatibility**: 最接近现有基础设施, 且复用面最大。目标不是证明 Picard scheme representability, 而是保证 *Picard functor* 的定义和 base change maps 没有方向错误。
2. **Descent of equality after algebraic closure**: 把 “after base change to $\bar{k}$ the diagram commutes, hence over k ” 抽成 reusable lemma/checklist。
3. **Cohomology/base change theorem registry**: 为每次 “by cohomology and base change” 生成 hypothesis checklist 与 Lean theorem stub, 先做审计, 不先做证明。
4. **Abelian variety rigidity / theorem of the cube / no rational curves**: 适合比较不同证明路线, 提取 properness、separatedness、connectedness、base point、field assumptions 等隐藏条件。
5. **Heavy computation microbenchmarks**: 从 Chern class、Hilbert polynomial、dimension count、generating function 中抽出可由 Lean algebra 证明的真实计算任务。

1.4 本报告的结论一句话

最有希望的研究产出不是 “一个形式化了某篇名著的巨大 Lean project”, 而是一套 **source-grounded, hypothesis-aware, reusable formal audit task library**: 每个 task 有论文出处、原始 proof smell、数学 statement 草案、Lean imports、依赖等级、隐藏假设 checklist 和复用目标。

2 核心价值定位

2.1 这不是完整形式化论文

本方向的工作对象不是 “论文”, 而是论文中的 *technical obligations*。一个 obligation 可能只是一句 “the construction is compatible with base change”, 也可能是一个 diagram chase、一个 spectral sequence index convention、一个 quotient functor 的 well-definedness、一个 descent datum 的 cocycle condition, 或者一个 Chern class calculation。形式化目标可以是以下任一层次:

1. **Hypothesis audit**: 列出被引用 theorem 所需的 hypotheses, 并检查论文是否逐项满足。
2. **Lean-style statement**: 把 obligation 写成 Lean theorem signature 或 blueprint statement; 证明可以是 sorry, 但 hypotheses 必须精确。
3. **Partial proof / abstract proof**: 在抽象 category/group/module/scheme property 层面证明可复用 lemma。
4. **Full local formalization**: 只对一个小 lemma 完成 Lean proof, 例如 quotient well-definedness、pullback-tensor coherence、descent of equality、polynomial identity。

这与传统 “formalize paper” 不同。完整形式化通常追求从定义到主定理的端到端闭环; 本方向追求的是把真实论文中最容易隐藏错误的 technical glue 抽成高复用、低颗粒度、可审计的基础设施任务。

2.2 六类价值及优先级

2.2.1 重计算 verify outsourcing

优先级: 中高。重计算论文中的 intersection calculation、Chern class calculation、Hilbert polynomial、dimension count、generating function、spectral sequence book-keeping 往往很适合 Lean/Mathlib 的代数、组合、线性代数、formal power series 能力。但如果计算依赖完整 Chow ring 或 cohomology ring presentation, 则前置基础设施会很重。因此推荐先剥离 “计算核心”: 把几何 theorem 降维为 polynomial identity、finite sum identity、graded degree check 或 filtration indexing check。

2.2.2 Hidden hypothesis audit

优先级: 最高。这是最符合 “低声望高风险” 的价值点。真实论文中经常出现:

by base change, by descent, standard, well-known, compatible with pullback, the diagram commutes, after extending the ground field.

这些语句很少错在主想法, 但很容易错在 hypotheses: Noetherian vs finite presentation、flatness、properness、separatedness、geometric connectedness、perfect field、separable

extension、cohomological flatness、base change map 的方向、sheafification 的 topology。Lean 的优势不是自动理解所有 EGA/SGA，而是迫使 obligation 的输入和输出变成类型级约束。

2.2.3 Black-box theorem interface specification

优先级：最高。对 FGA/EGA/SGA/Stacks Project 的大机器，不建议一开始形式化证明，而应先形式化接口。例如：

- *Picard functor representability*: 输入是 proper flat finite presentation 等 hypotheses，输出是 scheme/algebraic space/group object 及 universal property。
- *Quot scheme representability*: 输入是 projective/finitely presented/coherent/flat 条件，输出是 representing object 与 universal quotient。
- *Cohomology and base change*: 输入是 properness、coherence、flatness 或 perfectness、surjectivity/fiberwise condition，输出是 base change map isomorphism。
- *Artin representability*: 输入是一张 condition checklist，而不是一句 “by Artin’s criterion”。

接口规格本身就有学术价值：它能告诉读者“这篇论文到底调用了哪个黑箱版本”。

2.2.4 Reusable formal infrastructure extraction

优先级：最高。一个好的 audit task 不应该只服务于一篇论文。它应该能迁移到其他论文。例如：

- pullback of tensor product 的 naturality 可用于 Picard functor、Poincare bundle、Abel-Jacobi map、Fourier-Mukai kernel；
- descent of equality 可用于 algebraic closure arguments、Néron mapping property、Albanese universal property；
- Hilbert polynomial constancy 可用于 Hilbert/Quot、moduli of sheaves、compactified Picard；
- cohomology/base change map naturality 可用于 Picard scheme、seesaw theorem、relative duality、deformation theory。

2.2.5 Real-math benchmark for AI theorem proving

优先级：高。已有 benchmark 往往来自竞赛题、教材题、或 Mathlib 中已有 sorry。本方向可以补一个缺口：来自真实论文 **proof obligations** 的 **ugly but meaningful tasks**。这些 tasks 可能没有漂亮定理名，但正是研究数学中最常见的 technical debt。一个 benchmark item 应包含：source paragraph、proof smell、informal statement、Lean statement、dependency imports、hidden hypotheses、expected proof strategy、reuse tags。

2.2.6 Proof audit / technical due diligence for mathematical papers

优先级：中高。即使最后没有完成 Lean proof，审计报告也有价值：它可以指出某篇论文在哪些步骤依赖了哪些黑箱、是否误用 base change、是否漏掉 characteristic assumptions、是否把 algebraically closed field 的结果直接 descent 到任意 field。这类似数学论文的 *technical due diligence*，特别适合在高复杂度 moduli/AG papers 中使用。

2.3 本方向的输出形态

推荐把项目输出分为四种层次：

Audit memo 对一篇论文的 proof smells 与 hidden hypotheses 进行逐项标注。

Lean statement file
每个 obligation 写成 theorem/definition/namespace，允许先用 sorry，但 hypothesis list 精确。

Reusable lemma PR
对成熟的小任务提交 Mathlib 或项目库，例如 quotient functoriality、descent of equality、tensor-pullback coherence。

Benchmark dataset
把 tasks 结构化为 YAML/JSON：source、statement、difficulty、expected imports、status、dependency risk。

3 论文/主题候选总览

3.1 筛选原则

本报告没有只列最著名论文，而是按“能否抽出局部 audit task”筛选。优先考虑：

1. 结果重要，或被后续文献频繁引用；
2. proof 中出现大量 *technical glue, bookkeeping, base change, descent, compatibility, representability*;
3. 能抽出小 lemma 或 theorem interface，而不是必须重建整篇论文；
4. 审计可能发现 hidden hypotheses，例如 Noetherian、finite presentation、flatness、properness、separatedness、geometrically connected/reduced、perfectness、separability、characteristic；
5. 审计产物能复用到 Lean/Mathlib 或 Jacobian formalization project；
6. dependency explosion 可控，或者即便依赖爆炸，也能产出高质量 interface audit。

评分维度如下。除 **Dependency Explosion Risk** 外，5 分代表更适合；**Dependency Explosion Risk** 的 5 分代表更危险、更不宜作为首个 formalization target。

Infrastructure Fit 与 Lean/Mathlib 或当前 Jacobian formalization 基础设施的贴合度。

Local Auditability 能否从论文中切出局部可检查任务。

Hidden Hypothesis Risk
是否容易漏掉 hypotheses 或误用 black-box theorem。

Bookkeeping Burden proof 中是否存在大量低声望高风险推导。

Reusability 形式化/审计产物能否复用到其他论文或 Mathlib。

Dependency Explosion Risk
一碰就需要重建半个 EGA/SGA/GIT/derived stacks 的风险。

Report Value 即使不完成形式化，是否能产出有价值的审计报告。

表 1: 候选论文/主题总览。方向代码: A Picard/Albanese/Jacobian; B Hilbert/Quot/representability; C cohomology and base change; D descent/base change; E rigidity; F heavy computation; G benchmark。

ID	Title / 主题	Authors	Year	Link	Dir.	Audit value / 为什么适合审计	Pri.
P01	The Picard scheme	Steven Kleiman	L. 2005	arXiv	A/C/D	Picard functor、fpqc sheafification、representability、base change hypotheses 的核心接口；适合拆成相对 Picard 的黑箱 specification 与局部 functoriality checks。	P0
P02	Compactifying the Picard scheme	Allen B. Altman, Steven L. Kleiman	1980	DOI	A/B	compactified Picard 的源头之一；大量 moduli functor、rank-one torsion-free sheaf、flatness/open condition 技术。	P1
P03	Compactifying the relative Jacobian over families of reduced curves	Eduardo Esteves	2001	DOI	A/D/B	relative compactified Jacobian 与 reduced curve family 接近 Jacobian formalization；适合审计 degree、stability、base change 与 fine/coarse moduli 的接口。	P1
P04	A compactification of the universal Picard variety over the moduli space of stable curves	Lucia Caporaso	1994	DOI	A/B/D	Universal Picard variety 的经典高引用构造；GIT + balanced line bundles + stable curves, technical glue 丰富。	P2
P05	Néron models and compactified Picard schemes over the moduli stack of stable curves	Lucia Caporaso	2005/07	arXiv	A/D	Néron model、compactified Picard 与 moduli stack 连接；大量 base change/descent 与 universal property 检查。	P1

ID	Title / 主题	Authors	Year	Link	Dir.	Audit value / 为什么适合审计	Pri.
P06	Compactified stacks over the moduli stack of stable curves with marked points	Picard Margarida Melo	2008/11	arXiv	A/B/D	Picard stacks over moduli stack 的 compactification; 适合作为 stacky descent 与 functorial compactified Picard 的压力测试。	P2
P07	Autoduality of compactified Jacobians for curves with plane singularities	Dima Arinkin	2010/13	arXiv	A/F	Poincare sheaf extension、Fourier-Mukai、Cohen-Macaulay sheaf 的技术风险很高; 适合作为 black-box interface + local compatibility audit。	P1
P08	Fourier-Mukai and autoduality for compactified Jacobians I/II	Margarida Melo, Antonio Rapagnetta, Filippo Viviani	2012/13	arXiv arXiv II	I, A/F	推广 Arinkin 类型的 autoduality; 技术 obligation 密集, 适合抽取 Poincare bundle、base change、connected component 等任务。	P2
P09	Construction of Hilbert and Quot Schemes	Nitin Nitsure	2005	arXiv	B/C	Hilbert/Quot representability 与 flattening stratification 的可读入口; 适合作为 Lean-style theorem interface 的模板。	P0
P10	Quot Functors for Deligne-Mumford Stacks	Martin Olsson, Jason Starr	2002	arXiv	B/D	stacky Quot 的 representability 结果, 说明“Quot exists”在 stack setting 下 hypotheses 多且易漏。	P1
P11	Cohomology and base change for algebraic stacks	Jack Hall	2012/17	arXiv	C/B	base change theorem 的 stack generalization; 适合将“cohomology and base change”黑箱拆成 precise hypotheses 与 base change map naturality。	P1
P12	Moduli of complexes on a proper morphism	Max Lieblich	2005/06	arXiv	B/C	derived moduli 的 Artin stack 代表性; 适合审计 Artin representability prerequisites 与 deformation-theory bookkeeping。	P2

ID	Title / 主题	Authors	Year	Link	Dir.	Audit value / 为什么适合审计	Pri.
P13	Versal deformations and algebraic stacks	Michael Artin	1974	DOI	B	Artin representability 的基础文本；不是直接形式化对象，而是 black-box theorem interface specification 的源头。	P2
P14	Algebraic approximation of structures over complete local rings	Michael Artin	1969	EuDML	B	Artin approximation 是大量 algebraization/representability proof 的黑箱；适合审计 “algebraize formal object” 的精确前提。	P3
P15	Moduli of representations of the fundamental group of a smooth projective variety II	Carlos T. Simpson	1994	DOI	B/C/F	Higgs/Betti/de Rham moduli 的经典文献；heavy GIT、Chern class、cohomology/deformation bookkeeping, 适合 negative/control benchmark。	P3
P16	On the moduli of vector bundles on an algebraic surface	David Gieseker	1977	DOI	B/F	经典 moduli of vector bundles; GIT、Hilbert polynomial、boundedness、semistability 技术适合审计接口，但依赖爆炸大。	P2
P17	Semistable sheaves in positive characteristic	Adrian Langer	2004	DOI	B/F/D	positive characteristic semistability 与 boundedness 结果常用于 moduli；hidden characteristic assumptions 风险高。	P2
P18	A complete answer to Albanese base change for incomplete varieties	Jeffrey D. Achter, Sebastian Casalain-Martin, Charles Vial	2022	arXiv	D/A	明确讨论 Albanese 在 field extension 下何时 base change；与 “pass to algebraic closure and descend back” 高度贴合。	P0
P19	Para-abelian varieties and Albanese maps	Bruno Laurent, Stefan Schröer	2021/24	arXiv , DOI	A/D/E	在非理想 base-field/fiber 条件下给出 Albanese maps；适合审计 imperfect fields、universal homeomorphisms 与 descent。	P1
P20	Albanese maps for open algebraic spaces	Stefan Schröer	2022	arXiv	D/A	open algebraic spaces 的 Albanese；functoriality、field extension、descent 情况丰富。	P1

ID	Title / 主题	Authors	Year	Link	Dir.	Audit value / 为什么适合审计	Pri.
P21	An intersection number for the punctual Hilbert scheme of a surface	Geir Ellingsrud, Arild Strømme	1996	arXiv	F	短而明确的 intersection number 结果; 适合作为 P2 重计算 verification benchmark。	
P22	Chern classes of tau-tological sheaves on Hilbert schemes of points on surfaces	Manfred Lehn	1999	DOI	F/C	Chern classes、tautological sheaves、Hilbert schemes 的算法化计算; 非常适合 long calculation 审计。	P2
P23	Tautological module and intersection theory on Hilbert schemes of nodal curves	Ziv Ran	2009	arXiv	F/A	relative Hilbert schemes of nodal curves 与 tau-tological module; 接近 curve/Jacobian 但计算重。	P3
P24	SorryDB: Can Provers Real-World Theorems?	AI Austin Letson et al. Lean	2026	arXiv	G	真实 Lean 项目中的 open tasks benchmark; 可作 P1 为本方向 benchmark 设计的参照, 但需要补足“来自论文 proof obligations”的维度。	

表 2: 候选项评分矩阵。除 Dependency Explosion Risk 外,5 分代表更适合;Dependency Explosion Risk 的 5 分代表风险更高、更容易“牵一发而动全身”。

ID	Fit	Local	Hidden	Bookkeeping	Reuse	Dep.Explosion	Report	一句理由
P01	5	5	5		5	5	3	5 Picard functor、fpqc sheafification、representability、base change hypotheses 的核心接口；适合拆成相对 Picard 的黑箱 specification 与局部 functoriality checks。
P02	4	4	5		5	5	4	5 compactified Picard 的源头之一；大量 moduli functor、rank-one torsion-free sheaf、flatness/open condition 技术。
P03	4	4	5		5	5	4	5 relative compactified Jacobian 与 reduced curve family 接近 Jacobian formalization；适合审计 degree、stability、base change 与 fine/coarse moduli 的接口。
P04	3	4	5		5	4	5	5 Universal Picard variety 的经典高引用构造；GIT + balanced line bundles + stable curves，technical glue 丰富。
P05	4	4	5		4	5	4	5 Néron model、compactified Picard 与 moduli stack 连接；大量 base change/descent 与 universal property 检查。
P06	3	4	5		5	4	5	4 Picard stacks over moduli stack 的 compactification；适合作为 stacky descent 与 functorial compactified Picard 的压力测试。
P07	3	4	5		5	5	5	5 Poincare sheaf extension、Fourier-Mukai、Cohen-Macaulay sheaf 的技术风险很高；适合作为 black-box interface + local compatibility audit。
P08	3	4	5		5	5	5	4 推广 Arinkin 类型的 autoduality；技术 obligation 密集，适合抽取 Poincare bundle、base change、connected component 等任务。
P09	4	5	5		5	5	4	5 Hilbert/Quot representability 与 flattening stratification 的可读入口；适合作为 Lean-style theorem interface 的模板。
P10	3	4	5		5	5	5	5 stacky Quot 的 representability 结果，说明“Quot exists”在 stack setting 下 hypotheses 多且易漏。
P11	3	4	5		5	5	5	5 base change theorem 的 stack generalization；适合将“cohomology and base change”黑箱拆成 precise hypotheses 与 base change map naturality。
P12	2	3	5		5	4	5	4 derived moduli 的 Artin stack 代表性；适合审计 Artin representability prerequisites 与 deformation-theory bookkeeping。
P13	2	3	5		4	5	5	5 Artin representability 的基础文本；不是直接形式化对象，而是 black-box theorem interface specification 的源头。
P14	1	2	5		3	4	5	3 Artin approximation 是大量 algebraization/representability proof 的黑箱；适合审计“algebraize formal object”的精确前提。
P15	2	3	5		5	4	5	4 Higgs/Betti/de Rham moduli 的经典文献；heavy GIT、Chern class、cohomology/deformation bookkeeping，适合 negative/control benchmark。
P16	2	4	5		5	4	5	4 经典 moduli of vector bundles；GIT、Hilbert polynomial、boundedness、semistability 技术适合审计接口，但依赖爆炸大。
P17	2	3	5		5	4	5	4 positive characteristic semistability 与 boundedness 结果常用于 moduli；hidden characteristic assumptions 风险高。

ID	Fit	Local	Hidden	Bookkeeping	Reuse	Dep.Explosion	Report	一句理由
P18	4	5	5		4	5	3	5 明确讨论 Albanese 在 field extension 下何时 base change; 与 “pass to algebraic closure and descend back” 高度贴合。
P19	3	4	5		4	4	4	5 在非理想 base-field/fiber 条件下给出 Albanese maps; 适合审计 imperfect fields、universal homeomorphisms 与 descent。
P20	3	4	5		4	4	4	5 open algebraic spaces 的 Albanese; functoriality、field extension、descent 情况丰富。
P21	3	5	3		5	4	3	4 短而明确的 intersection number 结果; 适合作为重计算 verification benchmark。
P22	3	5	4		5	4	4	5 Chern classes、tautological sheaves、Hilbert schemes 的算法化计算; 非常适合 long calculation 审计。
P23	3	4	4		5	3	5	3 relative Hilbert schemes of nodal curves 与 tautological module; 接近 curve/Jacobian 但计算重。
P24	5	5	2		4	5	2	5 真实 Lean 项目中的 open tasks benchmark; 可作为本方向 benchmark 设计的参照, 但需要补足 “来自论文 proof obligations” 的维度。

3.2 优先级粗分

- P0** 立即进入 deep read 或 statement extraction: Kleiman *The Picard scheme*, Nitsure *Construction of Hilbert and Quot Schemes*, Achter–Casalaina–Martin–Vial Albanese base change, Picard quotient/pullback/descent tasks, cohomology/base change theorem registry。
- P1** 作为第一批论文卡片深读: Altman–Kleiman、Esteves、Caporaso 2005、Arinkin、Olsson–Starr、Hall、Laurent–Schröer、Schröer、benchmark 设计文献。
- P2** 作为第二批或专项参考: Caporaso 1994、Melo、Gieseker、Langer、Lehn、Ellingsrud–Strømme。
- P3** 目前只作为 reference 或 negative/control example: Artin approximation full proof、Simpson/GIT/analytic-heavy papers、Ran 的相对 Hilbert scheme 交理论全貌。

4 分方向详细分析

4.1 A. Picard / Albanese / Jacobian / line bundles

4.1.1 数学背景

Picard/Jacobian formalization 的中心对象不是单个 Jacobian，而是一整套 line-bundle infrastructure: invertible sheaves、tensor product、pullback、relative Picard functor、Picard scheme、 Pic^0 、Poincaré bundle、degree map、Abel–Jacobi map、Albanese universal property、base change 与 descent。Stacks Project 明确把 Picard functor 作为 relative construction 讨论：普通 $X \mapsto \text{Pic}(X)$ 不可代表，但相对版本在适当条件下有代表性；Kleiman 的 *The Picard scheme* 则是可读的 FGA/Picard scheme 入口。

4.1.2 常见 proof smell

- “The Picard functor is represented by a scheme” 但未写 proper/flat/finitely presented/geometric fiber conditions。

- “The construction is compatible with base change” 但未说明是 raw functor、sheafification 后 functor，还是 representing object 的 base change。
- “The Poincare bundle is normalized” 但未说明 normalization 是沿哪一个 section，是否唯一到唯一同构。
- “This map factors through Pic^0 ” 但未检查 connected component、degree zero、geometrically connected/reduced assumptions。
- “The Abel-Jacobi map is functorial” 但 diagram 中 line bundle pullback/tensor inverse 的方向未逐项核对。

4.1.3 推荐候选论文

- Steven L. Kleiman, *The Picard scheme*, [arXiv:math/0504020](https://arxiv.org/abs/math/0504020).
- Altman–Kleiman, *Compactifying the Picard scheme*, [DOI:10.1016/0001-8708\(80\)90043-2](https://doi.org/10.1016/0001-8708(80)90043-2).
- Eduardo Esteves, *Compactifying the relative Jacobian over families of reduced curves*, [DOI:10.1090/S0002-9947-01-02746-5](https://doi.org/10.1090/S0002-9947-01-02746-5).
- Lucia Caporaso, *A compactification of the universal Picard variety over the moduli space of stable curves*, [DOI:10.1090/S0894-0347-1994-1254134-8](https://doi.org/10.1090/S0894-0347-1994-1254134-8).
- Dima Arinkin, *Autoduality of compactified Jacobians for curves with plane singularities*, [arXiv:1001.3868](https://arxiv.org/abs/1001.3868).

4.1.4 Lean/Mathlib 贴合度

高, 但需要分层。Mathlib 公开文档显示已有 scheme category、morphism properties、AG sites、modules over schemes、Kähler differentials、ring-level Picard group 等基础。不过 scheme-level line bundles、relative Picard functor、Picard scheme representability 和 Poincare bundle 需要项目库或新增基础。因此首批 Lean tasks 应该落在：

- pullback of tensor products 的 naturality;
- quotient by pullback line bundles 的 functoriality;
- relative Picard functor 的 theorem signature;

- descent of equality 或 line bundle uniqueness;
- Abel map $q \mapsto \mathcal{O}(q - p)$ 的 base change square。

4.1.5 依赖爆炸风险

Picard functor 的“定义层”可控; Picard scheme representability 会迅速依赖 Hilbert/Quot、cohomology/base change、flattening stratification、Artin/FGA。Compactified Jacobian 更会引入 torsion-free rank-one sheaves、stability、compactified moduli、singular curves。因此推荐：先形式化 **functoriality/coherence/descent**，后把 **representability** 当作 **black-box interface**。

4.1.6 最适合抽取的 Lean-style task

Task A1. Let $f : X \rightarrow Y$ be a morphism of schemes and let L, M be invertible sheaves on Y . Construct the canonical isomorphism

$$f^*(L \otimes_{\mathcal{O}_Y} M) \cong f^*L \otimes_{\mathcal{O}_X} f^*M$$

and prove its compatibility with associativity, unit, inverse line bundle and isomorphism classes in $\text{Pic}(Y)$.

这个 task 不是华丽 theorem, 但它会被 Picard functor、Poincare bundle、Abel-Jacobi map、Fourier-Mukai kernels 重复使用。

4.2 B. Hilbert / Quot / flattening / representability

4.2.1 数学背景

Hilbert scheme、Quot scheme 和 flattening stratification 是现代 moduli constructions 的基础。Nitsure 的讲义 *Construction of Hilbert and Quot Schemes* 是很好的 FGA-style 入口; Olsson–Starr 的 *Quot Functors for Deligne-Mumford Stacks* 展示了当对象从 schemes 推广到 stacks 后, representability 的输出可能变成 algebraic space, 且 quasi-projectivity 需要额外 hypotheses。Lieblich 的 *Moduli of complexes on a proper morphism* 则说明 derived moduli 的 Artin stack 代表性会引入 deformation theory 与 Grothendieck existence。

4.2.2 常见 proof smell

- “Choose the Hilbert/Quot scheme” 但未说明 projective/finitely presented/coherent/flat hypotheses。
- “By flattening stratification” 但未说明 universal property 的 index set 和 partial order。
- “The universal quotient pulls back” 但未检查 quotient equivalence relation 和 exactness。
- “The moduli functor is an algebraic stack by Artin’s criterion” 但未逐项验证 limit preservation、effectivity、openness、deformation/obstruction theory。

4.2.3 推荐候选论文

- Nitin Nitsure, *Construction of Hilbert and Quot Schemes*, [arXiv:math/0504590](https://arxiv.org/abs/math/0504590).
- Martin Olsson and Jason Starr, *Quot Functors for Deligne-Mumford Stacks*, [arXiv:math/0204307](https://arxiv.org/abs/math/0204307).
- Max Lieblich, *Moduli of complexes on a proper morphism*, [arXiv:math/0502198](https://arxiv.org/abs/math/0502198).
- Michael Artin, *Versal deformations and algebraic stacks*, [DOI:10.1007/BF01390174](https://doi.org/10.1007/BF01390174).

4.2.4 Lean/Mathlib 贴合度

Mathlib 的 scheme/morphism property/Proj 基础有帮助，但完整 Hilbert/Quot scheme construction 目前不应作为 first pilot。更现实的是做 theorem interface：

QuotRepresentabilityInterface: input = projective morphism $X \rightarrow S$, coherent/finitely presented sheaf $\mathcal{F}$, Hilbert polynomial P , flatness conditions; output = representing object $Quot_{\mathcal{F}/X/S}^P$, universal quotient, base-change compatibility.

4.2.5 依赖爆炸风险

很高。Hilbert/Quot 代表性本身会牵出 projective embedding、graded module regularity、flatness, coherent sheaves, Fitting ideals, noetherian hypotheses。Stack/derived 版本还会牵出 algebraic spaces/stacks 与 Artin representability。因此本方向的首要产出应该是 **black-box theorem interface specification** 与 **downstream citation audit**。

4.2.6 最适合抽取的 Lean-style task

Task B1. Given a quotient $\mathcal{F}_T \twoheadrightarrow \mathcal{G}$ in the Quot functor and a morphism $T' \rightarrow T$, construct the pulled-back quotient on $X_{T'}$ and prove functoriality under composition $T'' \rightarrow T' \rightarrow T$.

这个任务不需要先证明 Quot scheme 存在；它只需要把 Quot functor 的 raw functoriality 做对。

4.3 C. Cohomology and base change / higher direct image

4.3.1 数学背景

Cohomology and base change 是 Picard scheme、seesaw theorem、relative duality、deformation theory、moduli of sheaves 共同依赖的黑箱。Conrad notes、Stacks Project、Hall 的 algebraic stacks 版本都显示：不同 theorem 版本的 hypotheses 差异很大。在论文审计中，最危险的不是 theorem 不存在，而是作者引用了“看起来同名但 hypotheses/output 不同”的版本。

4.3.2 常见 proof smell

- “By cohomology and base change” 但未说明是 flat base change、proper base change、Grauert theorem 还是 derived projection formula。
- 没有检查 f proper、 $\mathcal{F}$ coherent、 $\mathcal{F}$ flat over base、base locally Noetherian/finitely presented。
- 把 surjectivity of a base change map at one fiber 误读为 neighborhood isomorphism，或忽略 dimensions of H^i locally constant。

- base change map 的方向写反, 或忘记 naturality under iterated base change。

4.3.3 推荐候选论文/资料

- Jack Hall, *Cohomology and base change for algebraic stacks*, [arXiv:1206.4179](https://arxiv.org/abs/1206.4179).
- Stacks Project, *Cohomology and base change, I/IV/V*, e.g. [Tag 02KE](#) and [Tag 08IM](#).
- Brian Conrad, *Cohomology and Base Change* notes, used here as a theorem-statement reference when publicly accessible.

4.3.4 Lean/Mathlib 贴合度

当前更适合做抽象 interface 与 naturality lemma, 而不适合做 full sheaf cohomology proof. Riou 的 Lean/mathlib derived category formalization 说明 homological algebra 基础在扩展, 但 AG sheaf cohomology 和 higher direct image 仍是大工程。因此推荐建立:

- theorem registry: 每个 theorem version 的 hypotheses/output;
- base change map 的 formal name 与方向;
- “paper citation -> theorem version” mapping;
- abstract naturality square proofs。

4.3.5 依赖爆炸风险

高。full proof 可能需要 derived categories of sheaves、K-injective/K-flat resolutions、quasi-coherent sheaves、properness finiteness theorem。但 theorem registry 和 naturality checks 具有高 report value。

4.3.6 最适合抽取的 Lean-style task

Task C1. In an abstract category-theoretic interface for a cartesian square, define a base change transformation $g^* R^i f_* \Rightarrow R^i f'_* g'^*$ and prove compatibility under composition of base changes.

即便 $R^i f_*$ 暂时是 black-box functor, 这个 task 也能捕获真实论文中常见的 coherence 风险。

4.4 D. Descent / base change / algebraic closure arguments

4.4.1 数学背景

“Pass to algebraic closure and descend back” 是代数几何论文中最常见、也最容易隐藏假设的动作之一。如果只需证明两个 morphisms 相等, faithfully flat descent of equality 可能足够; 如果要 descent line bundles、group schemes、Picard components、universal properties 或 representability, 则需要更多数据与 cocycle conditions。Achter–Casalaina–Martin–Vial 对 Albanese base change 的现代结果尤其提醒我们: separable extension 与 purely inseparable extension 的行为可能不同, perfect field assumption 不能随便省略。

4.4.2 常见 proof smell

- “It suffices to work over $\bar{k}$ ” 但没有说明最终 statement 是否 fpqc local。
- equality of maps 能 descent, 但 existence of object 或 universal property 未必能同样 descent。
- 忘记 separability/perfectness; 把 algebraic closure 当成 separable extension。
- base-changed object 有 canonical isomorphism, 但 descent datum 的 cocycle condition 未检查。
- connected component、identity component、 Pic^0 、maximal abelian subvariety 在 field extension 下不一定按 naive 方式 commute。

4.4.3 推荐候选论文

- Achter–Casalaina–Martin–Vial, *A complete answer to Albanese base change for incomplete varieties*, [arXiv:2210.05017](https://arxiv.org/abs/2210.05017).
- Bruno Laurent and Stefan Schröer, *Para-abelian varieties and Albanese maps*, [arXiv:2101.10829](https://arxiv.org/abs/2101.10829), DOI:10.1007/s00574-023-00378-0.
- Stefan Schröer, *Albanese maps for open algebraic spaces*, [arXiv:2204.02613](https://arxiv.org/abs/2204.02613).

- Stacks Project, *Descent*, [Stacks Project](#).

4.4.4 Lean/Mathlib 贴合度

较高, 尤其是 equality descent、properties stable under base change/local on target、separated diagonal 等抽象任务。Mathlib AG morphism properties 与 fpqc site 模块已有不少入口; 但 descent of line bundles、descent of schemes/group schemes/representability 仍需新增基础。

4.4.5 依赖爆炸风险

从 equality descent 到 object descent, 风险阶梯非常明显:

- equality of morphisms: 中等, 可优先做;
- descent of properties: 中等到高, 取决于 property 在 Mathlib 中的 local-on-target API;
- descent of line bundles: 高, 需要 sheaf/module descent;
- descent of group schemes/Picard schemes/Albanese universal property: 很高, 适合接口化。

4.4.6 最适合抽取的 Lean-style task

Task D1. Let $u : S' \rightarrow S$ be faithfully flat. If two morphisms $a, b : X \rightarrow Y$ over S have equal pullbacks to S' , then $a = b$, under hypotheses that make $\mathrm{Hom}_S(X, Y)$ an fpqc sheaf or that Y has separated diagonal.

这是许多 algebraic closure arguments 的最小核心。

4.5 E. Abelian variety rigidity / rational maps / theorem of the cube

4.5.1 数学背景

Rigidity Lemma、Theorem of the Cube、Theorem of the Square、Seesaw theorem、no rational curves in abelian varieties、rational maps into abelian varieties 等结果是 Picard/Albanese/Jacobian constructions 的经典工具。它们经常被当成“标准事实”使用，但 hypotheses 很细：properness、connectedness、geometric integrality、separatedness、base point、normality、regularity of rational maps、field assumptions。

4.5.2 常见 proof smell

- “By rigidity” 但没有说明 source proper/connected、target separated、fiberwise constant 条件。
- “Rational map to an abelian variety extends” 但未说明 source normal/proper 或 indeterminacy resolution。
- “There are no rational curves in an abelian variety” 但未说明 morphism vs rational map、field algebraic closedness、properness。
- “Theorem of the cube” 中 line bundle restrictions 与 tensor product 的 sign/convention 容易错。

4.5.3 推荐资料/论文类型

- Conrad/Milne abelian varieties notes: Rigidity theorem、rational maps into abelian varieties、theorem of cube 的可读入口。
- Stacks Project, *Theorem of the cube*, [Tag 0BEZ](#).
- Albanese/Picard duality 文献中使用 rigidity/cube/seesaw 的 proof steps。

4.5.4 Lean/Mathlib 贴合度

中等。Mathlib 有 `scheme/morphism properties`、`rational map` 模块与 `AG group` 模块入口；但 `abelian varieties` 的完整理论、`proper group schemes`、`Picard/dual abelian variety` 仍不完整。首批任务可绕开大对象，先做 *rigidity lemma interface* 与 *line bundle tensor restriction bookkeeping*。

4.5.5 依赖爆炸风险

`Rigidity lemma` 的抽象版本较可控；`theorem of cube` 需要 `line bundles on products`、`flat proper finite presentation`、`finite locally free modules` 等基础；`rational map extension` 会牵出 `normality/properness/resolution`，风险更高。

4.5.6 最适合抽取的 Lean-style task

Task E1. Encode the hypotheses of a rigidity lemma and prove the pure categorical consequence: if two maps to a separated target agree on a schematically dense/proper-connected family of fibers, then the induced factorization is unique.

这个版本可以先抽象化，避免一开始建立 `abelian variety` 的完整理论。

4.6 F. Heavy computation / intersection theory / spectral sequence bookkeeping

4.6.1 数学背景

有些论文的主要风险不是隐藏假设，而是长计算：`intersection number`、`Chern class`、`Hilbert polynomial`、`dimension count`、`spectral sequence page indexing`、`long exact sequence`、`generating function`、`operator algebra`。Lean 对这类任务的优势明显：一旦计算被抽成 `algebraic/combinatorial statement`，机器检查能避免符号、`index`、`degree shift` 错误。

4.6.2 推荐候选论文

- Ellingsrud–Strømme, *An intersection number for the punctual Hilbert scheme of a surface*, [arXiv:alg-geom/9603015](https://arxiv.org/abs/alg-geom/9603015).
- Manfred Lehn, *Chern classes of tautological sheaves on Hilbert schemes of points on surfaces*, DOI:10.1007/s002220050307, [arXiv:math/9803091](https://arxiv.org/abs/math/9803091).
- Ziv Ran, *Tautological module and intersection theory on Hilbert schemes of nodal curves*, [arXiv:0905.2229](https://arxiv.org/abs/0905.2229).
- Gieseker and Langer type moduli papers 中的 Hilbert polynomial/Chern class/stability inequality computations。

4.6.3 常见 proof smell

- “A straightforward calculation gives” 后面接一个符号复杂公式。
- spectral sequence 的 index convention 从 homological 变 cohomological, 符号改变。
- Chern class/character 的 normalization 不一致。
- dimension count 中少减一个 automorphism dimension 或 stabilizer dimension。
- Hilbert polynomial 的 leading coefficient/factorial convention 错位。

4.6.4 Lean/Mathlib 贴合度

如果目标是完整 intersection theory, 贴合度低; 如果目标是计算核心, 贴合度高。例如 polynomial identities、formal power series identities、integer inequalities、finite sums、matrix rank/dimension formulas、graded degree bookkeeping 都可立即尝试。

4.6.5 依赖爆炸风险

两层区分非常重要:

几何解释层	Hilbert schemes, Chow rings, cohomology operations, tautological bundles: 依赖爆炸高。
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计算核心层 polynomial/formal series/finite sum/degree checks: 依赖爆炸低, 适合 Lean pilot。

4.6.6 最适合抽取的 Lean-style task

Task F1. Translate a Chern class generating-function identity from Lehn-style calculation into a formal power series equality over a commutative ring, then prove it using Mathlib algebra/formal power series tools.

4.7 G. Real-math Lean benchmark extraction

4.7.1 数学背景

AI theorem proving 的 benchmark 正在快速变化。ProofNet 以教材级问题为主; LeanDojo 强调 premise selection 和 challenging split; SorryDB 从真实 Lean projects 中抽取 open tasks; ArXivLean 尝试从近期 arXiv 论文提取 research-level Lean statements。本方向的独特位置是: 从真实论文 **proof obligations** 中抽取小而丑的 **infrastructure tasks**, 而不是从主定理或摘要中抽题。

4.7.2 常见 benchmark 缺口

- 竞赛题 benchmark 过于漂亮, 不反映研究论文 technical glue。
- 真实 Lean project 的 sorry 未必有清楚论文出处。
- arXiv 主定理抽取过难, 常需要几千行新基础设施。
- 缺少 hidden hypothesis checklist 和 source paragraph。

4.7.3 推荐候选资料

- SorryDB: *Can AI Provers Complete Real-World Lean Theorems?*, [arXiv:2603.02668](#).
- LeanDojo: *AI-Assisted Theorem Proving in Lean*, [LeanDojo](#).
- ProofNet, [ProofNet repository](#).

- *ArXivLean*, [MathArena ArXivLean page](#).

4.7.4 Lean/Mathlib 贴合度

非常高，因为 benchmark schema 和 task metadata 可以立即建设，不依赖代数几何大定理完成。每个 task 可以有状态：

- statement-only;
- compiles-with-sorry;
- partial-proof;
- complete-proof;
- blocked-by-infrastructure。

4.7.5 最适合抽取的 Lean-style task

Task G1. Define a YAML/JSON schema for formal audit tasks: source paper, source paragraph, proof smell, informal statement, Lean statement, imports, hidden hypotheses, dependency risk, proof status, reuse tags, reviewer notes.

5 候选论文卡片

本节给出 18 个最有审计价值或最有代表性的候选项。每张卡片的 Priority score 采用上一节的七维评分顺序：Infrastructure Fit / Local Auditability / Hidden Hypothesis Risk / Bookkeeping Burden / Reusability / Dependency Explosion Risk / Report Value。

5.1 P01. *The Picard scheme*

Authors	Steven L. Kleiman
Year	2005
Link	arXiv:math/0504020

Main result in one sentence

系统整理 Grothendieck 所设想的 Picard scheme 理论, 解释 relative Picard functor 何时可代表。

Technical infrastructure used

Schemes, sheaves of modules, invertible sheaves, fpqc descent, connected fibers, cohomology/base change, representability interfaces.

Suspicious / high-risk proof steps

“Picard functor is a sheaf / representable / commutes with base change” 常被后续论文压成一句话; 需要检查 flat/proper/finitely presented/geometrically connected 等假设是否被继承。

Candidate formal audit tasks

定义 relative Picard functor 的 Lean-style interface。; 形式化 $\text{Pic}(X)/\text{Pic}(S)$ 的 functoriality 与 group-valued quotient。; 审计某个 base change square 的自然性。

Required Mathlib / Lean infrastructure

Mathlib 已有 schemes、morphism properties、sites、modules 基础; 需项目中 line bundles/Picard functor/fpqc sheafification glue。

Hidden assumptions to check

Noetherian vs finite presentation; 是否需要 section; 几何连通/几何约化; fpqc sheafification; representability 是 scheme/algebraic space/stack。

Reusability 极高; 几乎所有 Jacobian、Albanese、compactified Picard 论文都会复用。

Priority score

5/5/5/5/5/3/5; Priority P0

Recommendation

deep read + formalize one lemma.

5.2 P02. Compactifying the Picard scheme

Authors Allen B. Altman, Steven L. Kleiman

Year 1980

Link [DOI:10.1016/0001-8708\(80\)90043-2](https://doi.org/10.1016/0001-8708(80)90043-2)

Main result in one sentence

构造 Picard scheme 的紧化, 成为后续 compactified Jacobian 文献的重要基础。

Technical infrastructure used

Moduli functors, Hilbert/Quot, flatness, coherent sheaves, Fitting ideals, determinant-like conditions.

Suspicious / high-risk proof steps

“rank-one torsion-free” 在非平滑纤维上的稳定性、universal family 的存在与 descent、等价关系是否 stacky。

Candidate formal audit tasks

把 compactified Picard functor 抽成黑箱接口。; 检查 torsion-free rank-one property 是否在指定 base change 后保持。

Required Mathlib / Lean infrastructure

需大量 coherent sheaf 与 flat families 基础; Lean immediate proof 不现实, 但适合作接口审计。

Hidden assumptions to check

Base scheme locally Noetherian; fibers reduced/integral; flatness of families; boundedness/Quot 依赖。

Reusability 高; Esteves、Caporaso、Arinkin 等后续路线都会引用。

Priority score

4/4/5/5/5/4/5; Priority P1

Recommendation

deep read for interface, not full formalization.

5.3 P03. *Compactifying the relative Jacobian over families of reduced curves*

Authors Eduardo Esteves

Year 2001

Link [DOI:10.1090/S0002-9947-01-02746-5](https://doi.org/10.1090/S0002-9947-01-02746-5)

Main result in one sentence

在 reduced curve family 上构造 relative Jacobian 的紧化。

Technical infrastructure used

Families of curves, line bundles/torsion-free sheaves, degree maps, compactified Picard, Quot schemes.

Suspicious / high-risk proof steps

reduced vs integral, Gorenstein/planar singularities, fiberwise degree 与 total degree 的兼容。

Candidate formal audit tasks

formal audit: degree of rank-one torsion-free sheaf behaves under base change. ; stability condition 的 pullback invariance statement.

Required Mathlib / Lean infrastructure

需要项目中 line bundles、degree、families of curves、Quot/flatness 接口；Mathlib 当前不足。

Hidden assumptions to check

Reduced fibers, relative dimension one, flatness, polarization choice, fine moduli vs coarse moduli.

Reusability 高；compactified Jacobian 论文中的 routine checks 可复用。

Priority score

4/4/5/5/5/4/5；Priority P1

Recommendation

formalize one lemma after deep read.

5.4 P04. *A compactification of the universal Picard variety over the moduli space of stable curves*

Authors Lucia Caporaso

Year 1994

Link [DOI:10.1090/S0894-0347-1994-1254134-8](https://doi.org/10.1090/S0894-0347-1994-1254134-8)

Main result in one sentence

构造 stable curves moduli space 上 universal Picard variety 的紧化。

Technical infrastructure used

Stable curves, line bundles and multidegrees, Hilbert/GIT quotient, Picard stack interfaces.

Suspicious / high-risk proof steps

balanced condition 的 numerical inequalities、GIT quotient 与 moduli functor 的 compatibility、automorphism/descent 问题。

Candidate formal audit tasks

把 balanced inequality 抽成可验证的 combinatorial lemma。; 审计 quotient map 与 pullback line bundles 的 functoriality。

Required Mathlib / Lean infrastructure

GIT、stable curves、stacks 过重; 但 multidegree 计算可局部形式化。

Hidden assumptions to check

Characteristic、stability chamber、universal curve existence、automorphism groups、separatedness/properness。

Reusability 中高; 尤其适合 benchmark 的 numerical bookkeeping。

Priority score

3/4/5/5/4/5/5; Priority P2

Recommendation

keep as high-value reference; formalize combinatorial sublemmas.

5.5 P05. *Néron models and compactified Picard schemes over the moduli stack of stable curves*

Authors Lucia Caporaso

Year 2005/2007

Link [arXiv:math/0502171](https://arxiv.org/abs/math/0502171)

Main result in one sentence

比较 compactified Picard schemes 与 Néron models 在 stable curve

moduli 上的关系。

Technical infrastructure used

Relative Picard, Néron models, stable curves, moduli stacks, descent.

Suspicious / high-risk proof steps

“the same functor after base change” 与 Néron mapping property 的 domain restrictions; regular vs stable model.

Candidate formal audit tasks

Lean-style interface for Néron mapping property as uniqueness of extension。; 检查某个 comparison morphism 的 base change compatibility。

Required Mathlib / Lean infrastructure

Néron model formalization 基本缺失; 但 universal property/equality descent 可局部做。

Hidden assumptions to check

Regularity, smooth locus, trait/DVR hypotheses, automorphism stack, separatedness.

Reusability 高; Jacobians over families 与 degeneration 方向复用。

Priority score

4/4/5/4/5/4/5; Priority P1

Recommendation

deep read + extract universal-property obligations.

5.6 P06. *Compactified Picard stacks over the moduli stack of stable curves with marked points*

Authors Margarida Melo

Year 2008/2011

Link [arXiv:0811.0763](https://arxiv.org/abs/0811.0763)

Main result in one sentence

构造带标记稳定曲线模空间上的 compactified Picard stacks。

Technical infrastructure used

Algebraic stacks, stable curves, Picard stacks, balanced/stability data.

Suspicious / high-risk proof steps

stack representability, DM vs Artin, dimension count, marked point pullback compatibility。

Candidate formal audit tasks

审计 stack morphism 的 representability/smoothness hypotheses。; 将 degree/stability rule 抽成 Lean combinatorial task。

Required Mathlib / Lean infrastructure

Stacks infrastructure 远超当前 Mathlib; 可做外部接口规范和组合子任务。

Hidden assumptions to check

Genus restrictions, marked point sections, automorphisms, smoothness/irreducibility。

Reusability 中高; 对 compactified Picard/Jacobian 方向有用。

Priority score

3/4/5/5/4/5/4; Priority P2

Recommendation

keep as reference + extract benchmark tasks.

5.7 P07. *Autoduality of compactified Jacobians for curves with plane singularities*

Authors Dima Arinkin

Year 2010/2013

Link [arXiv:1001.3868](https://arxiv.org/abs/1001.3868)

Main result in one sentence

对具有 planar singularities 的 integral projective curves 证明 compactified Jacobian 的 autoduality。

Technical infrastructure used

Compactified Jacobian, Poincare sheaf, derived categories/Fourier-Mukai, Cohen-Macaulay sheaves.

Suspicious / high-risk proof steps

plane singularities、integral curve、flatness and CM conditions、kernel normalization; derived statement 方向容易错。

Candidate formal audit tasks

interface: Poincare sheaf extends as flat CM family。; formal audit: normalization uniqueness of Poincare line bundle。; 检查 autoduality morphism 的 functoriality square。

Required Mathlib / Lean infrastructure

Lean derived categories 有基础进展, 但 AG derived sheaves/FM 仍远; 适合接口规格。

Hidden assumptions to check

Planar singularities、integrality、projectivity、compactified Jacobian fine moduli、field assumptions。

Reusability 高; Poincare/Abel-Jacobi/Jacobian formalization 直接相关。

Priority score

3/4/5/5/5/5/5; Priority P1

Recommendation

deep read as stretch target; formalize only normalization/naturality sublemmas.

5.8 P08. *Fourier-Mukai and autoduality for compactified Jacobians I/II*

Authors Margarida Melo, Antonio Rapagnetta, Filippo Viviani

Year 2012/2013

Link [arXiv:1207.7233](https://arxiv.org/abs/1207.7233), [arXiv:1308.0564](https://arxiv.org/abs/1308.0564)

Main result in one sentence

研究 compactified Jacobians 的 Fourier-Mukai 变换与 autoduality。

Technical infrastructure used

Compactified Jacobian, generalized Jacobian, derived functors, sheaf kernels, base change.

Suspicious / high-risk proof steps

connected component identification、base change of compactified Jacobian、kernel support/coherence。

Candidate formal audit tasks

审计 generalized Jacobian 到 Pic^0 的 connected component map。; 定义 kernel composition 的 black-box statement。

Required Mathlib / Lean infrastructure

当前不适合完整证明; Lean task 应限于 Picard/Jacobian interface 和 line bundle naturality。

Hidden assumptions to check

Singularity assumptions、polarization choices、connectedness/reducedness、derived boundedness。

Reusability 高但起步成本高。

Priority score

3/4/5/5/5/5/4; Priority P2

Recommendation

keep as reference; formalize one compatibility square only.

5.9 P09. Construction of Hilbert and Quot Schemes

Authors Nitin Nitsure

Year 2005

Link [arXiv:math/0504590](https://arxiv.org/abs/math/0504590)

Main result in one sentence

以 FGA 路线讲解 Hilbert 与 Quot schemes 的 Grothendieck 构造。

Technical infrastructure used

Projective morphisms, coherent sheaves, flatness, Hilbert polynomials, flattening stratifications.

Suspicious / high-risk proof steps

flattening stratification 的 universal property、regularity bound、base-change of quotients 常被作为标准引理引用。

Candidate formal audit tasks

formal interface for Quot functor base change。 ; verify flatness locus universal property as statement stub。 ; Hilbert polynomial constancy in a flat proper family。

Required Mathlib / Lean infrastructure

Mathlib 项目中可从 morphism properties/Proj/affine schemes 起步； Hilbert/Quot scheme 本身需大量新基础。

Hidden assumptions to check

Noetherian/projective/finitely presented； very ample line bundle； flatness vs constant Hilbert polynomial。

Reusability 极高；所有 moduli construction 论文复用。

Priority score

4/5/5/5/5/4/5； Priority P0

Recommendation

deep read + interface formalization.

5.10 P10. Quot Functors for Deligne-Mumford Stacks

Authors Martin Olsson, Jason Starr

Year 2002

Link [arXiv:math/0204307](https://arxiv.org/abs/math/0204307)

Main result in one sentence

证明 DM stack 上的 Quot functor 可由 separated locally finitely presented algebraic space 表示。

Technical infrastructure used

Deligne-Mumford stacks, algebraic spaces, Quot functor, proper support.

Suspicious / high-risk proof steps

论文中的“under additional hypotheses”与后续引用中常被抹平；quasi-projectivity 不等于 representability。

Candidate formal audit tasks

记录 Quot functor representability 的 exact hypothesis schema。；审计某篇后续论文是否把 quasi-projective conclusion 错用为 general。

Required Mathlib / Lean infrastructure

Stacks/algebraic spaces beyond Mathlib；但适合 report-level interface audit。

Hidden assumptions to check

Separatedness、locally finitely presented、proper support、base S 是否 noetherian。

Reusability 高；moduli of sheaves on stacks 方向复用。

Priority score

3/4/5/5/5/5/5；Priority P1

Recommendation

deep read at interface level.

5.11 P11. Cohomology and base change for algebraic stacks

Authors Jack Hall

Year 2012/2017

Link [arXiv:1206.4179](https://arxiv.org/abs/1206.4179)

Main result in one sentence

证明 algebraic stacks 的 cohomology and base change 以及相关 Ext functor coherence。

Technical infrastructure used

Derived categories, coherent sheaves, algebraic stacks, base change maps.

Suspicious / high-risk proof steps

不同版本的 base change theorem 对 noetherian/proper/flat/perfect/tor-amplitude 依赖不同，后续论文容易引用错版本。

Candidate formal audit tasks

创建 cohomology/base change theorem registry。; 形式化 base change map 的 naturality square (先作 abstract category statement)。

Required Mathlib / Lean infrastructure

Lean derived category 已有基础，但 sheaf cohomology/higher direct image on schemes/stacks 仍大量缺失。

Hidden assumptions to check

Properness, finite presentation, flatness/perfectness, coherence, tameness、derived boundedness。

Reusability 极高; Picard、Hilbert、moduli 都依赖。

Priority score

3/4/5/5/5/5/5; Priority P1

Recommendation

interface specification + one abstract naturality lemma.

5.12 P12. *Moduli of complexes on a proper morphism*

Authors Max Lieblich

Year 2005/2006

Link [arXiv:math/0502198](https://arxiv.org/abs/math/0502198)

Main result in one sentence

在 proper morphism 下证明一大类 derived category objects 形成 locally finitely presented Artin stack。

Technical infrastructure used

Derived categories, t-structures, Artin stacks, deformation theory.

Suspicious / high-risk proof steps

flatness in derived hearts、boundedness/perfectness、automorphism stack、limit preservation。

Candidate formal audit tasks

Artin representability checklist as machine-readable interface。; 抽取 S-flat coherent sheaves 子情形的 precise statement。

Required Mathlib / Lean infrastructure

极重; Lean 现阶段只适合做接口规格和部分 category/homological algebra。

Hidden assumptions to check

Proper morphism、flatness、finite presentation、deformation functor conditions、openness。

Reusability 中高; moduli of sheaves/complexes 方向基础。

Priority score

2/3/5/5/4/5/4; Priority P2

Recommendation

only keep as reference unless derived infrastructure is target.

5.13 P13. Versal deformations and algebraic stacks

Authors Michael Artin

Year 1974

Link [DOI:10.1007/BF01390174](https://doi.org/10.1007/BF01390174)

Main result in one sentence

发展 versal deformation 与 algebraic stacks 的代表性理论。

Technical infrastructure used

Formal moduli, algebraic stacks, deformation theory, algebraization.

Suspicious / high-risk proof steps

现代论文经常写 “by Artin’s criterion” 但不列出各项 condition; 每项 condition 都可能隐藏 technical debt。

Candidate formal audit tasks

制定 Artin criterion hypothesis checklist。; 审计目标论文中哪一项 condition 被引用但未证明。

Required Mathlib / Lean infrastructure

当前不适合 Lean proof; 适合自然语言 +Lean-like specification。

Hidden assumptions to check

Effectivity、openness、automorphisms、limit preserving、complete local rings。

Reusability 极高; moduli paper audit 核心。

Priority score

2/3/5/4/5/5/5; Priority P2

Recommendation

interface checklist, not formal proof.

5.14 P16. *On the moduli of vector bundles on an algebraic surface*

Authors David Gieseker

Year 1977

Link [DOI:10.2307/1971157](https://doi.org/10.2307/1971157)

Main result in one sentence

研究代数曲面上向量丛的模空间构造。

Technical infrastructure used

Vector bundles on surfaces, GIT, Hilbert schemes, Chern classes.

Suspicious / high-risk proof steps

大量 numerical inequalities 和 stability bookkeeping, 可切出组合/多项式任务。

Candidate formal audit tasks

验证某个 Hilbert polynomial transformation.; 抽取 stability inequality 的 algebraic manipulation task。

Required Mathlib / Lean infrastructure

GIT 不现实; 计算子任务可用 algebra/ring tactics 做。

Hidden assumptions to check

Polarization、smooth projective surface、boundedness assumptions、characteristic。

Reusability 中；moduli/GIT 计算任务有 benchmark 价值。

Priority score

2/4/5/5/4/5/4；Priority P2

Recommendation

formalize computation only.

5.15 P18. *A complete answer to Albanese base change for incomplete varieties*

Authors Jeffrey D. Achter, Sebastian Casalaina-Martin, Charles Vial

Year 2022

Link [arXiv:2210.05017](https://arxiv.org/abs/2210.05017)

Main result in one sentence

给出 incomplete varieties 的 Albanese base change 问题的完整答案, 并说明 purely inseparable 的修正。

Technical infrastructure used

Albanese varieties, universal property, field extensions, descent/base change.

Suspicious / high-risk proof steps

“Albanese commutes with base change” 在非完备或非 separable extension 下可能有修正项；非常适合 hidden hypothesis audit。

Candidate formal audit tasks

形式化 separable field extension 下 universal property 的 transport。；构造 checklist: pass to algebraic closure 是否只需 separable descent。

Required Mathlib / Lean infrastructure

Abelian varieties/Albanese scheme 当前不完整；universal property/equality descent 可先抽象化。

Hidden assumptions to check

Separable vs purely inseparable, perfect field, regularity/smoothness/incomplete varieties.

Reusability 高; Albanese/Jacobian base change 论文常见风险。

Priority score

4/5/5/4/5/3/5; Priority P0

Recommendation

deep read + formalize abstract universal-property lemma.

5.16 P19. *Para-abelian varieties and Albanese maps*

Authors Bruno Laurent, Stefan Schröer

Year 2021/2024

Link [arXiv:2101.10829](https://arxiv.org/abs/2101.10829), [DOI:10.1007/s00574-023-00378-0](https://doi.org/10.1007/s00574-023-00378-0)

Main result in one sentence

对 proper algebraic spaces 构造到 para-abelian varieties 的 Albanese map, 并给出相对与 base-change 形式。

Technical infrastructure used

Algebraic spaces, group schemes, Stein factorization, Albanese universal property.

Suspicious / high-risk proof steps

imperfect fields, 缺少 rational points/ample sheaves, non-reduced/reducible 情形会打破常见简化。

Candidate formal audit tasks

审计 universal property 在 universal homeomorphism 下的转移。; 建立 base-field hypothesis checklist.

Required Mathlib / Lean infrastructure

Algebraic spaces/para-abelian 未在 Mathlib 完备; 但 universal-property task 可抽象化。

Hidden assumptions to check

Proper/flat/finitely presented, cohomologically flat in degree 0,

$h^0(O_{\{X_s\}})=1$, Pic^τ 的 maximal abelian subvarieties, imperfect field.

Reusability 中高; 帮助避免 classical theorem 的误用。

Priority score

3/4/5/4/4/4/5; Priority P1

Recommendation

deep read at hypothesis level.

5.17 P20. *Albanese maps for open algebraic spaces*

Authors Stefan Schröer

Year 2022

Link [arXiv:2204.02613](https://arxiv.org/abs/2204.02613)

Main result in one sentence

构造 open algebraic spaces 的 Albanese maps 并研究 functoriality/base change。

Technical infrastructure used

Algebraic spaces, affinization, para-abelian varieties, base change.

Suspicious / high-risk proof steps

“commutes with field extensions” 需要加 “up to universal homeomorphisms” 等限定; 易被后续引用弱化。

Candidate formal audit tasks

提取 base change theorem 的 exact statement。; formal audit: equality of two maps descends along faithfully flat cover。

Required Mathlib / Lean infrastructure

AG/algebraic spaces 过重; 可在 category/universal property 层面建模。

Hidden assumptions to check

Separated finite type、connected/reduced affinization、field imperfection。

Reusability 中高。

Priority score

3/4/5/4/4/4/5; Priority P1

Recommendation

hypothesis audit + abstract task.

5.18 P22. *Chern classes of tautological sheaves on Hilbert schemes of points on surfaces***Authors** Manfred Lehn**Year** 1999**Link** [DOI:10.1007/s002220050307](https://doi.org/10.1007/s002220050307)**Main result in one sentence**

给出 Hilbert schemes of points on surfaces 上 tautological sheaves Chern classes 的算法描述。

Technical infrastructure used

Chern classes, cohomology rings, Hilbert schemes, combinatorics/generating functions.

Suspicious / high-risk proof steps

index convention、sign、degree shift、operator algebra bookkeeping 非常容易错。

Candidate formal audit tasks

抽取 generating function identity 由 Lean 验证。; 检查 operator commutator signs 与 degree shifts。

Required Mathlib / Lean infrastructure

Cohomology/Chow theory 不全; 形式幂级数/多项式 identities 可先做。

Hidden assumptions to check

Smooth projective surface、characteristic/field assumptions、cohomology theory choice。

Reusability 中高; heavy-computation benchmark 非常好。

Priority score

3/5/4/5/4/4/5; Priority P2

Recommendation

formalize one algebraic calculation.

6 候选 formal audit tasks 清单

6.1 任务粒度原则

一个合格的 *formal audit task* 不应只是“形式化某篇论文的一部分”。它应满足：

1. **Source-grounded**: 有明确论文、章节、定理或 proof paragraph 来源。
2. **Hypothesis-explicit**: 列出所有 noetherian, fp, flat, proper, separated, geometric, perfect, characteristic, coherence hypotheses。
3. **Lean-shaped**: 可以被写成 Lean theorem signature、definition、namespace 或 blueprint theorem。
4. **Reusable**: 完成后能用于不止一篇论文。
5. **Dependency-aware**: 标注 readiness 1–4, 避免把 full EGA/SGA 当成首个任务。

表 3: 候选 formal audit tasks 清单。Readiness: 1 可立即/接近 Mathlib; 2 项目雏形或路线图; 3 需新增大量基础; 4 暂不适合形式化审计。

ID	Task	Source	Statement draft	Required infrastructure	Diff.	Reuse	Pri.	Readiness
T01	Relative Picard functor interface specification	P01, Stacks Project card functor, P03-P08	给定 $f: X \rightarrow S$, 声明 $\text{Pic}_{\{X/S\}}(T) = \text{Pic}(X_T)/\text{Pic}(T)$ 或其 fpqc sheafification; 记录 representability theorem 的 precise hypotheses 与 output category.	Schemes, functor category, Picard group/line bundles, fpqc sheafification; project Picard functor roadmap.	M	极高	P0	2 项目雏形/路线图
T02	Pullback-tensor compatibility for line bundles	P01, P03-P08, all card/Jacobian tasks	For a morphism and two invertible sheaves on the target, construct the canonical pullback-tensor isomorphism and prove naturality plus associativity/unit coherence.	Sheaves of modules, pullback, tensor product, invertible sheaves; category-theoretic coherence.	M	极高	P0	1/2 Mathlib + project glue
T03	Quotient by pullback line bundles is functorial	P01, Stacks functor	Picard If relative Picard is defined by quotienting by line bundles pulled back from the test scheme, verify pullback preserves that subgroup and induces a functor to Ab.	Groups, quotient groups, pullback compatibility, fiber products.	S-M	极高	P0	1/2
T04	Base change map for relative Picard	P01, P03-P08	Under hypotheses H on a family X over S and a base change $S' \rightarrow S$, construct the canonical relative-Picard base-change map and specify when it is an isomorphism.	Relative functors, sheafification, base change, representability interfaces.	H	极高	P0	2/3
T05	Descent of equality of morphisms along faithfully flat cover	P05, P18-P20, descent papers	de If $u: S' \rightarrow S$ is faithfully flat and two S-morphisms $a, b: X \rightarrow Y$ become equal after base change to S' , then $a = b$, under Y separated or via sheaf condition on Hom.	Fpqc topology/descent, morphism equality, separated diagonal or sheaf condition.	M	极高	P0	1/2
T06	Descent of a line bundle with cocycle data	P01, P03-P08, descent	Stacks A line bundle on an fpqc cover with cocycle data descends to X; verify cocycle, uniqueness, and tensor compatibility.	Fpqc descent for modules/line bundles, H sheafification, cocycle diagrams.	H	极高	P1	2/3
T07	Poincare bundle normalization uniqueness	P07-P08, P01	Given two Poincare line bundles normalized along a section, prove they are uniquely isomorphic; track tensoring by pullbacks from the base.	Line bundles, sections, pullback, quotient by base line bundles.	M-H	极高	P0	2
T08	Abel-Jacobi/Abel map functoriality square	P03-P08, Jacobian formalization roadmap	For a family of curves C/S with section p, define $q \mapsto O(q-p)$ and prove compatibility under base change $S' \rightarrow S$.	Relative curves, divisors/line bundles, H pullback, Picard functor.	H	极高	P1	2/3
T09	Seesaw theorem interface	P01, P19, theorem-of-cube ecosystem	Stacks For a proper flat morphism with suitable connected-fiber hypothesis, a line bundle trivial/pulled back fiberwise is pulled back from the base; specify exact hypotheses.	Cohomology/base change, Picard functor, line bundles.	H	极高	P1	2/3
T10	Theorem of the cube hypothesis audit	Stacks Project theorem of the cube, abelian variety literature	Encode a theorem-of-the-cube interface with flat/proper/finitely presented hypotheses and verify the restriction/naturality maps in a special line-bundle case.	Products/fiber products, line bundles, restrictions, tensor, finite locally free modules.	H	高	P1	2/3
T11	Rigidity Lemma interface and separatedness check	Abelian variety rigidity, P18-P20	rigid- If X is proper/connected over k and Y is separated, a morphism $X \times S \rightarrow Y$ constant on one fiber factors through S; list exact hypotheses and formalize equality descent core.	Properness, separatedness, fiber products, morphism equality.	M-H	高	P1	1/2
T12	No rational curves in an abelian variety	Abelian variety rigidity, Milne/Conrad notes	rigid- Every morphism $P^1_k \rightarrow A$ to an abelian variety is constant; compare proofs via rigidity lemma vs theorem of cube.	Projective line, abelian varieties/group schemes, rigidity lemma.	H	中高	P2	2/3
T13	Quot functor base-change compatibility	P09-P10	A quotient $q: F_T \rightarrow G$ flat over T pulls back to a quotient over T' ; verify equivalence relation and functoriality.	Sheaves/modules, pullback exactness, flatness, quotients, fiber products.	H	极高	P1	2/3
T14	Flattening stratification universal property	P09, Stacks flattening stratifications	flattening Given finite presentation module F on X over S, specify the locally closed stratum S_P such that $T \rightarrow S$ factors through S_P iff F_T is flat with Hilbert polynomial P.	Flatness, Fitting ideals, Hilbert polynomial, universal property.	H+	极高	P2	3
T15	Hilbert polynomial constancy in flat projective families	P09, P16	For F flat over S on projective X/S, fiber Hilbert polynomial is locally constant under noetherian/fp hypotheses.	Proj/projective spectrum, coherent sheaves, flatness, Hilbert polynomial.	H	高	P2	3

ID	Task	Source	Statement draft	Required infrastructure	Diff.	Reuse	Pri.	Readiness
T16	Cohomology/base change map naturality	P11, Stacks cohomology/base change	Construct the higher-direct-image base-change map and prove functoriality for composable base changes in an abstract interface.	Derived category/cohomology interface, H		极高	P0	2/3
T17	Grauert/cohomology base change hypothesis checklist	P11, Conrad notes, Stacks	Given a paper proof using “by cohomology and base change”, verify properness, flatness, finite presentation/noetherian, coherence, fiberwise dimension condition.	Mostly report/specification; optional Lean theorem stub with hypotheses.	M	极高	P0	1 report / 3 proof
T18	Long exact sequence diagram chase	C/F directions; moduli/cohomology papers	Formalize a diagram chase from a short exact sequence of sheaves to a rank/dimension equality or vanishing statement in an abstract abelian category.	CategoryTheory, homological algebra, M		高	P1	1
T19	Spectral sequence bookkeeping task	F direction, P15/P22	Given a first quadrant spectral sequence with specified vanishing region, verify the claimed edge isomorphism or filtration length.	Spectral sequence interface or custom finite-grid combinatorics.	H	中高	P2	2/3
T20	Chern class / generating function calculation	P21-P22, P16-P17	Translate a Chern class/generating function identity into a polynomial/formal power series equality and verify with Lean algebra.	Polynomial, formal power series, graded algebra; optional Chern-class interface.	M	高	P1	1
T21	Degree and multidegree bookkeeping on stable curves	P03-P06	For a nodal/stable curve decomposed into components, verify balanced/multidegree inequalities under normalization or base change.	Finite graphs, integer inequalities, divisors/multidegree data.	M	高	P1	1
T22	Stability condition invariant under base change	P03-P06, P16-P17	If a line bundle/sheaf satisfies a fiberwise stability or balanced condition over S, then after base change $S' \rightarrow S$ it satisfies the pulled-back condition.	Definitions of stability as predicates, base change of fibers, numerical inequalities.	M-H	高	P1	2
T23	Descent of properties: proper/flat/separated/fp	D direction; descent	Build a checklist and Lean lemmas for properties stable under base change and local on the target for fpqc topology.	Mathlib morphism properties, fpqc cover, descent instances.	M	极高	P0	1
T24	Albanese universal property under separable base change	P18-P20	Abstractly, if Alb_X represents morphisms from X to abelian/parabelian varieties, prove base-changed object represents the corresponding functor after separable extension, assuming target category base changes well.	Category/universal property, field extension, base-change functor.	M-H	高	P1	2
T25	Purely inseparable caveat checker	P18-P20	Given a theorem statement involving algebraic closure or ground-field extension, flag whether it uses separability/perfectness; require separate branch for purely inseparable extensions.	Metadata/report tool; optional Lean typeclass for separable/perfect field.	S	高	P0	1
T26	Poincare sheaf extension black-box theorem	as P07-P08	Specify a theorem object: under planar/integral/projective curve assumptions, a normalized Poincare line bundle on the smooth locus extends to a flat Cohen-Macaulay sheaf on compactified Jacobian product.	Compactified Jacobian interface, sheaves, flatness, line bundle normalization.	CM H+	高	P2	3/4
T27	Sheafification-pullback compatibility square	com-P01, all tasks	For a presheaf construction F, verify that sheafification followed by pullback is canonically isomorphic to pullback followed by sheafification in the required topology, with coherence.	Sheafification, sites, pullback functors, natural isomorphisms.	H	极高	P1	2/3
T28	Benchmark metadata schema for formal audit tasks	P24, LeanDojo, ProofNet, ArXivLean	Create a structured schema: source citation, informal statement, Lean imports, dependencies, hidden hypotheses, expected statement, proof status, downstream reuse.	Lean project tooling, YAML/JSON, Mathlib search, CI.	S-M	极高	P0	1

6.2 首批最推荐的 10 个 tasks

1. **T02 Pullback-tensor compatibility for line bundles**: 直接服务 Picard functor、Poincare bundle、Abel-Jacobi map。若 line bundles as invertible sheaves 已有项目雏形，这是最值得先做的 coherence task。
2. **T03 Quotient by pullback line bundles is functorial**: 这是 relative Picard functor 的 raw functoriality 核心，不需要先证明 representability。
3. **T05 Descent of equality of morphisms along faithfully flat cover**: 可覆盖大量 “over $\bar{k}$ checked, hence over k ” 的最小核心。
4. **T16 Cohomology/base change map naturality**: 即使 $R^i f_*$ 暂为 black-box functor，也能先证明 compositional coherence。
5. **T17 Grauert/cohomology base change hypothesis checklist**: 适合快速产生审计报告价值，能马上用于论文阅读。
6. **T23 Descent of properties: proper/flat/separated/fp**: 与 Mathlib morphism property API 贴合，复用面极大。
7. **T07 Poincare bundle normalization uniqueness**: 和 Jacobian/compactified Jacobian formalization 最贴近，目标小但意义大。
8. **T21 Degree and multidegree bookkeeping on stable curves**: 可用 Lean 的整数不等式/图论/组合能力先行，不依赖完整 compactified Picard。
9. **T20 Chern class / generating function calculation**: 从 heavy-computation 论文抽可验证代数核心，适合 AI theorem proving benchmark。
10. **T28 Benchmark metadata schema**: 不需要等数学基础设施完成即可启动；能让后续所有 tasks 都可检索、可复现、可评测。

6.3 示例 Lean-style statement 草案

以下草案并非最终 Lean 代码，而是用于说明如何把论文中的一句 “routine” 改写成形式化任务。

6.3.1 Picard quotient functoriality

Given a cartesian square induced by $u : T' \rightarrow T$ and $X_T = X \times_S T$, define

$$u_X^* : \text{Pic}(X_T) \rightarrow \text{Pic}(X_{T'})$$

and prove:

$$u_X^*(\text{im}(\text{Pic}(T) \rightarrow \text{Pic}(X_T))) \subset \text{im}(\text{Pic}(T') \rightarrow \text{Pic}(X_{T'})).$$

Therefore pullback induces a group homomorphism

$$\text{Pic}(X_T)/\text{Pic}(T) \rightarrow \text{Pic}(X_{T'})/\text{Pic}(T').$$

Hidden hypotheses: only those needed to define pullback/tensor/quotient on Picard groups; representability is irrelevant. This is why it is a good early task.

6.3.2 Descent of equality

Let $u : S' \rightarrow S$ be faithfully flat and $Y \rightarrow S$ separated. For S -morphisms $a, b : X \rightarrow Y$, if $a_{S'} = b_{S'}$, then $a = b$.

Depending on Mathlib APIs, this can be formulated via diagonal monomorphism, via sheaf condition on Hom, or via an abstract faithful functor induced by fpqc cover. The audit value is high because many algebraic-closure arguments only require this exact lemma, not full object descent.

6.3.3 Cohomology/base change theorem registry item

Record a theorem entry:

`cohomology_base_change_proper_flat_coherent`

Inputs: $f : X \rightarrow S$ proper, finitely presented, $\mathcal{F}$ coherent and flat over S , S locally noetherian or finite presentation version, and a fiberwise condition.

Output: canonical base change map is an isomorphism in a neighborhood / after arbitrary base change, depending on theorem version.

This is not a proof; it is a typed interface. The audit question is whether a paper's citation matches this exact theorem version.

6.3.4 Heavy computation benchmark

From a Chern class or Hilbert polynomial calculation, isolate the algebraic identity:

$$\sum_i (-1)^i \binom{n}{i} P(i) = Q(n)$$

or a formal power series equality. Then prove it in Lean using polynomial/formal power series infrastructure.

The benchmark should retain source metadata so the task is still a research-paper audit, not a synthetic exercise.

7 优先级路线图

7.1 阶段一：快速调研和筛选

目标	从 30–50 个候选论文/主题中筛到 10–20 个高价值候选。
输入	英文关键词、arXiv/DOI/Stacks Project/Mathlib docs、已有 Jacobian formalization roadmap、用户已有基础设施清单。
动作	1. 按七维评分给每篇论文打分； 2. 标注 proof smells: <i>by base change, by descent, standard, compatible, the diagram commutes</i>; 3. 初步判定 readiness: Mathlib-supported / project scaffold / major infrastructure / not suitable; 4. 过滤掉 conceptual insight 大于 technical glue、或 dependency explosion 与复用价值严重不匹配的论文。
输出	候选论文表、方向分类、初步 priority P0–P3。
成功标准	至少 10 个候选具有明确 local audit obligations; 至少 5 个候选能与当前 Lean/Mathlib 或 Jacobian formalization roadmap 直接挂钩。

7.2 阶段二：深度阅读和局部 statement 提取

目标	对 5–8 篇论文做 deep read, 抽出 20–40 个 candidate audit tasks。
输入	阶段一的 P0/P1 候选: 建议从 Kleiman、Nitsure、Achter–Casalaina–Martin–Vial、Caporaso 2005、Arinkin、Hall、Laurent–Schröer 中选择。
动作	1. 对每篇论文建立 proof smell log; 2. 每遇到 black-box theorem 引用, 记录 theorem name、source、hypotheses、output、paper 是否满足; 3. 每遇到 diagram/naturality/base change/descent, 写出 informal statement; 4. 将 statement 改写为 Lean-style theorem signature; 5. 给每个 task 标注 imports/readiness/dependency risk/reuse tags。
输出	20–40 个 task cards; 每个 task 包含 source paragraph、statement draft、hidden hypotheses checklist、Lean infrastructure requirement、priority。
成功标准	至少 10 个 tasks 属于 readiness 1 或 2; 至少 5 个 tasks 可在不证明 representability 的情况下推进。

7.3 阶段三：形式化试点

目标	选择 3–5 个局部 tasks 尝试 Lean statement / partial proof / blueprint。
推荐试点	1. Picard quotient functoriality: statement-only 到 partial proof; 2. Pullback-tensor naturality for line bundles: 若 line bundle API 已稳定, 尝试完整 proof; 3. Descent of equality after faithfully flat base change: 以 fpqc sheaf/Hom 或 separated diagonal 版本表达;

4. **Cohomology/base change theorem registry**: 以 Lean constants/typeclasses 表达黑箱接口;
5. **Heavy computation identity**: 选一个 Chern class/Hilbert polynomial 计算核心完成 proof。

输入	阶段二 task cards、项目当前 Mathlib import graph、已有 Jacobian blueprint。
输出	Lean files: Audit/Picard.lean、Audit/Descent.lean、Audit/BaseChange.lean、Audit/Benchmarks.lean 等。
成功标准	1. 至少 3 个 Lean statements 能 compile; 2. 至少 1 个 task 完成 full proof; 3. 至少 1 个 task 产出 theorem interface registry; 4. 至少 1 个 task 可作为 AI theorem proving benchmark item。

7.4 建议的 8 周执行节奏

周次	目标	交付物
1-2	完成 候选论文池 扩展与 P0/P1 深读选择	30-50 个候选条目; 10-20 个精筛候选; source links verified
3-4	深读三条 P0 线: Picard、Quot、Albanese base change	proof smell log; 10-15 个 task cards; black-box theorem registry 初稿
5-6	深读 2-3 篇高风险案例: Arinkin、Caporaso、Hall 或 Laurent-Schröer	追加 10-20 个 task cards; hidden hypothesis checklist
7	Lean statement sprint	3-5 个 compiling statements; 1 个计算 benchmark proof attempt
8	审计报告与 benchmark package	PDF audit memo; Lean blueprint; YAML/JSON benchmark schema; next PR plan

7.5 阶段性产物的文件组织建议

```
FormalAudit/  
  Audit/  
    Picard/  
      Functoriality.lean  
      PoincareNormalization.lean  
      AbelJacobiBaseChange.lean  
    Descent/  
      Equality.lean  
      Properties.lean  
    BaseChange/  
      CohomologyRegistry.lean  
      Naturality.lean  
    Benchmarks/  
      PolynomialIdentities.lean  
      Multidegree.lean  
  data/  
    papers.yaml  
    tasks.yaml  
    sources.bib  
  blueprint/  
    picard_functor.md  
    albanese_base_change.md  
    cohomology_base_change_registry.md
```

8 风险与反例

8.1 需要完整重建 EGA/SGA 的论文

有些论文 proof 中每一步都建立在 EGA/SGA/FGA 大机器上, 且没有清晰可切分的 local obligations。如果目标是完成 Lean proof, 这类论文不适合作为第一阶段。典型风险包括:

- full representability of Picard scheme;

- full Hilbert/Quot construction;
- full Artin representability criterion;
- full flattening stratification;
- full relative duality;
- full derived algebraic stacks。

这些内容仍可做 *interface specification*, 但不应承诺短期 formal proof。

8.2 Conceptual insight 大于 technical glue 的论文

有些论文的难点是概念构造、几何直觉、发现新的 correspondence 或新的 strategy, 而不是 bookkeeping。这种论文即使很重要, 也不一定适合“低声望高风险审计”。评估标准是: 能否从 proof 中抽出可复用 technical obligation。如果只能抽出“理解主想法”, 不合作 formal audit benchmark。

8.3 Analytic / transcendental methods 太多

例如某些 Hodge theory、nonabelian Hodge、complex analytic moduli、period map 文献, 可能包含大量 analytic comparison、Sobolev estimates、PDE、harmonic metrics。这些方向即使有 hidden hypotheses, 也与当前 Lean/Mathlib algebraic geometry/Jacobian infrastructure 贴合度较低。Simpson 相关文献可作为 high-risk reference, 但不宜作为 first pilot。

8.4 结论重要但局部不可切分

有些论文每个 lemma 都依赖主构造本身, 无法隔离出小 task。例如 full Fourier-Mukai equivalence on compactified Jacobians 需要 derived categories of coherent sheaves、compactified moduli、kernel convolution、Cohen-Macaulay sheaves、dualizing complexes。Arinkin/Melo-Rapagnetta-Viviani 很有价值, 但早期任务应只抽 Poincare normalization、base change of kernels、connected component comparison, 而不是 full equivalence。

8.5 依赖大量尚不存在基础设施且复用价值不明确

不推荐在第一阶段投入以下目标：

- 完整 GIT quotient construction;
- 完整 Chow ring/intersection theory formalization, 只为验证一篇论文的一个数字;
- 完整 algebraic stacks in Lean, 只为重现一个 Artin representability proof;
- 完整 Néron models, 只为某个 comparison theorem。

如果这些基础设施本身已有独立项目目标，则可以并行；否则 formal audit 应选择更小的接口化任务。

8.6 隐藏假设审计的假阳性风险

形式化审计很容易把作者省略的“领域内默认假设”误判为漏洞。报告应区分：

Actual gap 某个 theorem 的必要 hypothesis 明显没有满足，且没有替代证明。

Omitted but standard
论文未写，但上下文、引用或 standing assumptions 实际覆盖。

Ambiguous theorem version
论文引用名称不精确，需要确认所用版本。

Documentation debt
数学可能正确，但 proof 对读者不可审计；形式化接口有价值。

本方向的目标主要是减少 *documentation debt* 和 *ambiguous theorem version*，而不是把所有省略都解读为错误。

8.7 Lean 当前能力被高估的风险

报告中的“可形式化”并不等于“Mathlib 今天已经有所有定义”。更准确的分层是：

- 可立即检查：纯代数/组合/quotient/naturality/equality descent 的抽象版本；
- 需要项目基础设施：line bundles as sheaves、relative Picard functor、Abel-Jacobi blueprint；

- 需要大量 **Mathlib**: cohomology/base change、Hilbert/Quot、Picard scheme representability;
- 目前不现实: full GIT、full Artin representability、full derived stack/Fourier-Mukai proof。

9 最终推荐

9.1 最推荐优先探索的 5 个主题方向

9.1.1 1. Relative Picard functor and line-bundle coherence

推荐理由: 与 Jacobian formalization 最贴合, 局部 task 多, 复用价值极高, 且不必先证明 representability。

候选论文/资料:

- Kleiman, *The Picard scheme*.
- Stacks Project, *The Picard functor*.
- Altman–Kleiman, *Compactifying the Picard scheme*.
- Esteves, *Compactifying the relative Jacobian over families of reduced curves*.
- Arinkin, *Autoduality of compactified Jacobians for curves with plane singularities*.

首批 tasks: T01–T04, T07, T08, T27。

建议首个 Lean file: Audit/Picard/Functoriality.lean。先做 quotient functoriality 和 pullback-tensor compatibility; representability 全部作为 interface。

9.1.2 2. Descent of equality and algebraic-closure arguments

推荐理由: hidden hypothesis 风险高, 早期 Lean 证明相对可控, 且对 Albanese/Jacobian/base change 文献复用极高。

候选论文/资料:

- Achter–Casalaina-Martin–Vial, *A complete answer to Albanese base change for incomplete varieties*.
- Laurent–Schröer, *Para-abelian varieties and Albanese maps*.
- Schröer, *Albanese maps for open algebraic spaces*.
- Stacks Project descent chapters.

首批 **tasks**: T05, T06, T23, T24, T25。

建议首个 **Lean file**: `Audit/Descent/Equality.lean`。从 equality descent 开始, 不先做 full descent of objects。

9.1.3 3. Cohomology and base change theorem registry

推荐理由: 几乎所有 moduli/Picard/Jacobian 论文都会用, 但 full proof 当前不现实; 做 theorem registry 和 hypothesis checklist 的 report value 很高。

候选论文/资料:

- Hall, *Cohomology and base change for algebraic stacks*.
- Stacks Project, *Cohomology and base change*.
- Conrad notes on cohomology and base change.
- Picard scheme/Seesaw theorem applications.

首批 **tasks**: T09, T16, T17。

建议首个 **Lean file**: `Audit/BaseChange/CohomologyRegistry.lean`。把 theorem versions 写成 constants/typeclass interfaces, 附带 source links 和 hypotheses。

9.1.4 4. Hilbert/Quot/flattening as black-box interfaces

推荐理由: 所有 moduli construction 的底座; 依赖爆炸高, 但 interface audit 价值极大。

候选论文/资料:

- Nitsure, *Construction of Hilbert and Quot Schemes*.
- Olsson–Starr, *Quot Functors for Deligne-Mumford Stacks*.

- Lieblich, *Moduli of complexes on a proper morphism*.
- Artin, *Versal deformations and algebraic stacks*.

首批 **tasks**: T13, T14, T15, 以及 Artin criterion checklist。

建议首个 **Lean file**: `Audit/Representability/QuotInterface.lean`。不证明 Quot scheme 存在, 只定义 functoriality 与 representability theorem signature。

9.1.5 5. Heavy-computation microbenchmarks

推荐理由: 可快速产出完整 Lean proofs 或 AI theorem proving benchmark, 不依赖完整代数几何基础设施。

候选论文/资料:

- Ellingsrud–Strømme, *An intersection number for the punctual Hilbert scheme of a surface*.
- Lehn, *Chern classes of tautological sheaves on Hilbert schemes of points on surfaces*.
- Ran, *Tautological module and intersection theory on Hilbert schemes of nodal curves*.
- Caporaso/Gieseker/Langer 文献中的 multidegree、Hilbert polynomial、stability inequality 计算。

首批 **tasks**: T20, T21, T18, T19。

建议首个 **Lean file**: `Audit/Benchmarks/PolynomialIdentities.lean`。每个 benchmark 保留 source paragraph, 避免变成无出处的 synthetic exercise。

9.2 Recommended next-step plan

1. 先做一个 **Picard functor mini-audit**。选 Kleiman + Stacks Project Picard functor 作为 source, 写出 5 个 Lean-style statements: pullback-tensor compatibility, quotient functoriality, base change map, Poincare normalization uniqueness, Pic^0 hypothesis checklist。

2. 并行做一个 **descent mini-audit**。选 Achter–Casalaina-Martin–Vial 与 Laurent–Schröer, 专门记录 separable/perfect/purely inseparable 条件。目标是形成 “algebraic closure argument checklist”。
3. 建立 **theorem registry**。对 Picard representability、Hilbert/Quot representability、cohomology/base change、theorem of cube、seesaw、rigidity lemma 各写一个 interface card。
4. 选择 3 个 **Lean pilot**。建议顺序：
 - (a) quotient by pullback line bundles is functorial;
 - (b) descent of morphism equality after fpqc/faithfully flat cover;
 - (c) one polynomial/formal power series identity from heavy-computation paper。
5. 把 **tasks** 数据化。用 YAML/JSON 表示每个 task 的 source、statement、hypotheses、imports、dependency risk、proof status。这一步会让项目自然成为 AI theorem proving benchmark。

9.3 最终一句推荐

第一篇深读应选 **Kleiman, *The Picard scheme***; 第一组 Lean statements 应选 **Picard functoriality + line-bundle pullback/tensor coherence**; 第一个 hidden-hypothesis case study 应选 **Albanese base change and descent from algebraic closure**; 第一个 benchmark pilot 应选 **Chern class/Hilbert polynomial/generating-function calculation**。

10 附录：三轮调研与文稿修订日志

为避免一次性罗列文献，本报告按三轮进行了筛选和修订。

第一轮：英文关键词检索与候选池建立

检索重点包括：

- *Picard functor representability base change descent*;

- *relative Picard scheme cohomology and base change*;
- *Albanese variety universal property Picard scheme*;
- *Quot scheme flattening stratification representability*;
- *faithfully flat descent algebraic geometry hidden assumptions*;
- *rigidity lemma abelian variety rational maps*;
- *theorem of the cube Picard scheme Albanese*;
- *intersection theory computation verification algebraic geometry*.

第一轮得到约 60 个来源，按 Picard/Jacobian、Hilbert/Quot、cohomology/base change、descent/Albanese、rigidity、heavy computation、AI benchmark 七类粗分。

第二轮：评分与依赖风险重估

第二轮将候选项压缩为 24 个总览条目和 18 张详细论文卡片。评分时特别下调了以下类别的 formalization priority:

- full Artin representability;
- full Hilbert/Quot construction;
- full Fourier-Mukai equivalence on compactified Jacobians;
- full GIT construction;
- full Chow/intersection theory.

这些对象仍保留为 *black-box interface specification* 或 *report-level audit* 候选，但不作为第一批 Lean proof target。

第三轮：文稿修订与任务抽象

第三轮重点把论文层面的候选转化为 28 个 formal audit tasks，并按 readiness 1–4 标注。修订原则是：

1. 把“完整形式化论文”的表述全部改为“局部形式化审计”；
2. 对 Mathlib 当前能力采用保守表述；

- 3. 将 Picard/line bundle/descent/cohomology-base-change/Hilbert-Quot/heavy-computation 分别区分为 proof target、statement target、interface target 或 report-only target;
- 4. 所有候选论文表格保留公开链接或 DOI/arXiv 链接;
- 5. 对 unclear 或 dependency-heavy 对象标注 uncertainty 或 not recommended for first pilots。

最终报告的建议是：第一阶段先产生审计报告和 Lean statement files；第二阶段才选择少量可证明 lemmas；不要把 FGA/SGA 大定理的完整形式化当作默认目标。

11 附录：主要资料与链接

本附录列出报告中实际采用的主要公开链接。链接以 arXiv、DOI/journal page、Stacks Project、Mathlib 官方文档为主；对于只作为背景参考的资料，报告不把其作为独立候选论文。

Mathlib / Lean / formalization infrastructure

Item	Link / 说明
Lean 4 theorem prover	DOI
Mathlib GitHub	leanprover-community/mathlib4
Mathlib documentation index	Mathlib docs index
Mathlib Scheme docs	Scheme docs
Mathlib AG modules/morphism properties/sites	Mathlib docs index; AG modules list
Mathlib Modules.Tilde	Modules.Tilde docs
Mathlib Kähler differentials	RingTheory.Kaehler.Basic
Mathlib ring-level Picard group	RingTheory.PicardGroup
Schemes in Lean	DOI
Formalising the Proj Construction in Lean	DOI

Derived categories in DOI
Lean/mathlib

Stacks Project / standard references

Item	Link / 说明
Stacks Project, Picard functor	Tag 0D24
Stacks Project, flattening stratifications	Tag 052F
Stacks Project, cohomology and base change I	Tag 02KE
Stacks Project, cohomology and base change IV	Tag 08IM
Stacks Project, theorem of the cube	Tag 0BEZ
Conrad, applications of base change for coherent cohomology	PDF handout
Conrad/Tony Feng notes, abelian varieties	PDF notes
Milne, Abelian Varieties notes	AV.pdf

候选论文与主题链接

Authors	Title / 主题	Link
Kleiman	<i>The Picard scheme</i>	arXiv:math/0504020
Altman–Kleiman	<i>Compactifying the Picard scheme</i>	DOI
Esteves	<i>Compactifying the relative Jacobian over families of reduced curves</i>	DOI
Caporaso	<i>A compactification of the universal Picard variety over the moduli space of stable curves</i>	DOI
Caporaso	<i>Néron models and compactified Picard schemes over the moduli stack of stable curves</i>	arXiv:math/0502171

Melo	<i>Compactified Picard stacks over the moduli stack of stable curves with marked points</i>	arXiv:0811.0763
Arinkin	<i>Autoduality of compactified Jacobians for curves with plane singularities</i>	arXiv:1001.3868
Melo–Rapagnetta–Viviani	<i>Fourier-Mukai and autoduality for compactified Jacobians I/II</i>	arXiv:1207.7233 ; arXiv:1308.0564
Nitsure	<i>Construction of Hilbert and Quot Schemes</i>	arXiv:math/0504590
Olsson–Starr	<i>Quot Functors for Deligne-Mumford Stacks</i>	arXiv:math/0204307
Hall	<i>Cohomology and base change for algebraic stacks</i>	arXiv:1206.4179
Lieblich	<i>Moduli of complexes on a proper morphism</i>	arXiv:math/0502198
Artin	<i>Versal deformations and algebraic stacks</i>	DOI
Artin	<i>Algebraic approximation of structures over complete local rings</i>	EuDML
Simpson	<i>Moduli of representations of the fundamental group of a smooth projective variety II</i>	DOI
Gieseker	<i>On the moduli of vector bundles on an algebraic surface</i>	DOI
Langer	<i>Semistable sheaves in positive characteristic</i>	DOI
Achter–Casalaina-Martin–Vial	<i>A complete answer to Albanese base change for incomplete varieties</i>	arXiv:2210.05017
Laurent–Schröer	<i>Para-abelian varieties and Albanese maps</i>	arXiv:2101.10829 ; DOI
Schröer	<i>Albanese maps for open algebraic spaces</i>	arXiv:2204.02613
Ellingsrud–Strømme	<i>An intersection number for the punctual Hilbert scheme of a surface</i>	arXiv:alg-geom/9603015
Lehn	<i>Chern classes of tautological sheaves on Hilbert schemes of points on surfaces</i>	DOI ; arXiv:math/9803091
Ran	<i>Tautological module and intersection theory on Hilbert schemes of nodal curves</i>	arXiv:0905.2229

Letson et al.

SorryDB: Can AI Provers Complete [arXiv:2603.02668](https://arxiv.org/abs/2603.02668)
Real-World Lean Theorems?

不确定性说明

- 本报告没有声称上述论文中的具体步骤存在错误；“风险”指的是适合审计、hypotheses 容易被省略或误配。
- 对 Conrad notes、Milne notes 等讲义资料，本报告仅作为 theorem-statement/reference background，不把它们作为待审计研究论文。
- 对 full GIT、full Artin representability、full derived stacks、full Fourier-Mukai equivalence，本报告明确标记为不适合首批形式化审计。